TEACHING VOCABULARY DEVELOPMENT USING INTERACTIVE WHITEBOARDS TO THIRD GRADE ELL STUDENTS IN THE UNITED ARAB EMIRATES

By

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A DISSERTATION PRESENTED TO THE GRADUATE SCHOOL
OF THE UNIVERSITY OF FLORIDA IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
DOCTOR OF EDUCATION

UNIVERSITY OF FLORIDA

2015

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To my wife, Ginger, and my daughter Marie for helping me on this journey

ACKNOWLEDGMENTS

I want to thank my wife, Ginger, for her love, support and endurance throughout this dissertation process. Without her, it would have been difficult to complete. I want to thank my daughter, Marie, for her love and understanding. I also want to thank my family for encouraging me and providing support for the completion of this dissertation.

I thank my advisor and chair of my committee, Dr. Swapna Kumar, for her support throughout the entire program and particularly during this dissertation process. Without her, I would not have made it this far. She has a passion for research and teaching students, and she has inspired me to achieve what seemed impossible to achieve. I am incredibly thankful for her support and mentorship.

I also would like to thank my co-chair, Dr. Maria Coady, for her advice and feedback throughout this process. She was incredibly helpful with guiding me through the data analysis phase, and I am thankful for all of her support. I also thank the other members of my committee, Dr. Ana Puig and Dr. Kara Dawson, for their encouragement and support during the many stages of my study.

I would also like to acknowledge the teachers and students at my school in Abu Dhabi for their participation. I thank them for their feedback and willingness to participate in this study. I also thank all of my colleagues in Cohort 2 of the Educational Technology program for their support.

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Abstract of Dissertation Presented to the Graduate School
of the University of Florida in Partial Fulfillment of the
Requirements for the Degree of Doctor of Education

TEACHING VOCABULARY DEVELOPMENT USING INTERACTIVE WHITEBOARDS
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By

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August 2015

Chair: Swapna Kumar
Cochair: Maria Coady
Major: Curriculum and Instruction

This research examined the use of interactive whiteboards (IWB) for English as a second language student vocabulary development. Assisting students with their vocabulary development was one of the goals of my district and the purpose of the study was to investigate how the use IWB materials can assist in this area. Participants included two native English-speaking teachers and four classes from a local elementary school for boys in Abu Dhabi. The study examined the development and implementation of IWB-based lessons.

The study used exploratory, qualitative methods for collecting and analyzing data within the framework of comprehensible input and Second Language Acquisition theory (SLA). Data collection methods included observations of classrooms, teacher interviews, a reflective journal, and recorded screenshots of the lesson. Thematic analysis (Braun & Clarke, 2006) was used for analyzing the data in rich detail.

The findings revealed several multimodal features of IWBs that are useful for ELL student vocabulary development. Information was provided about improving my professional practice and implications for instructional design and schools. This work is

significant in that it provides implications for this specific environment as well as for instructional design of IWB based lessons. This study adds to the research about the use of IWBs for ELL students, comprehensible input, and vocabulary development.

CHAPTER 1
INTRODUCTION

The city of Abu Dhabi is in the United Arab Emirates (UAE). Abu Dhabi is the second largest city in the country and also one of the wealthiest cities in the world (Makrakis, 2005). The Abu Dhabi Education Council (ADEC), which is in control of the schools in the city, is pushing toward the integration of technology in all of the public schools (Abu Dhabi Education Council, 2013; Al Mekhlafi, 2006a). Beginning in 2011, interactive whiteboards and computers were installed in all public elementary schools, and the schools have received a directive to deliver instruction using this technology. Consequently, professional development has been conducted to train teachers to use this technology. An interactive whiteboard (IWB) is a large display that is connected to a computer that allows a user to control the computer using a pen, finger, or other device. My school has been specifically instructed to use the installed IWBs for teaching and learning. This study explores the use of the IWB for teaching English as a second language in elementary classrooms in the UAE using materials designed according to prior research on IWBs and vocabulary instruction.

During this research project I was a third grade teacher and a practitioner researcher in an all-boys elementary school in Abu Dhabi. I have previous experience teaching second, fourth, fifth and 7th grades. I also had experience working as a Head of Faculty. I had received several sessions of professional development on using IWBs to teach vocabulary lessons for English Language Learner (ELL) students in the elementary school in which I teach. The students that I worked with were primarily local Emirati students and were classified as ELL students because they were in the process of learning English in addition to Arabic. English was used as a medium of instruction in

their classes, and English was spoken as a common language among the majority of the population (Gardner, 2001). Moreover, the majority of elementary public schools in Abu Dhabi employed native English speakers who taught English, mathematics, and science in their native language, English.

My School and ELL Instruction

The school in which I worked consisted of 85 percent students of Emirati origin. The remaining 15 percent of the students were mostly from Arabic countries such as Jordan, Egypt, and Syria, with a small minority coming from Western or Eastern European countries. The students had 35 instructional periods of classes each five-day school week, 15 of which were English, mathematics, and science. Each of the periods were 45 minutes in length. Math and science were taught in English by native English speakers and follow a prescribed curriculum. English class, which was five periods per week, was different in that the teachers were responsible for identifying and adapting curriculum and materials to use in the classroom. The remainder of the periods (i.e., 15 per week) consisted of Arabic/Islamic studies, social studies, physical education, art, music, and computer studies. Each of the classrooms in my school were equipped with two computers and an IWB.

The English language teaching approach that was used at my school was balanced literacy. Balanced literacy incorporates instruction in speaking, listening, reading and writing. Since there was a time limit each day, teaching all areas of literacy was difficult. This problem was alleviated by cross curricular planning and integrating reading and writing into one unit of instruction. The program that was used at my school for English development was PM Writing©. PM Writing© scaffolds English instruction for students through the use of modeled, shared, and guided writing, which encouraged

daily English vocabulary practice (Nelley, Croft, Smith, & Ciuffetelli, 2007). The program was integrated into the curriculum through whole class, small group, and independent reading and writing tasks for students.

Students were exposed to different text genres through writing exemplars. During this process, teachers modeled different reading and writing processes, drawing attention to text structures and the features of language used in the text. The teachers then modeled writing through usage of graphic organizers. Students then moved onto shared, guided, and independent reading activities, as well as guided and independent writing activities. Formal and informal assessments were given after each lesson and consisted of self-assessment, writing assessment tasks, and student writing samples. PM Writing© included an IWB component, which utilized interactive projectable books, writing exemplars, and various graphic organizers. This program was also supplemented through the use of other software such as Matching Game Maker©, Sentence Game Maker©, Microsoft PowerPoint©, and Easy Interactive Tools© (Languages Online, 2006).

Problem of Practice

Although many ELL students perform well in school, much of the research and data indicate that those students struggle with basic and academic content vocabulary (Echevarria & Graves, 2007). Students at my school struggled with the development of their English vocabulary, as evidenced by their External Measurement of Student Achievement (EMSA) scores (ADEC, 2013). Several factors appeared to be associated with these low scores, including lack of background knowledge or cognitive ability; however, the primary reason for low test scores may be due to limited vocabulary in English (Borgia, 2009; Wallace, 2007). Even though ELL students may have a large

vocabulary in their native language, the lack of vocabulary in the second language (L2, English) hinders academic achievement (Blachowicz, Fisher, & Ogle, 2006). By the time students reach high school, they do not have the English vocabulary necessary to attend university; therefore, they need remediation at that level (ADEC, 2013). The lack of vocabulary at the elementary level affects the students as they reach the middle, high, and university levels. This was a large problem in Abu Dhabi and the UAE. Given the availability of the IWBs in primary [elementary] schools in the UAE and the emphasis on the use of the IWBs, investigating the design and implementation of IWB based materials to facilitate vocabulary development is timely.

Prior research has demonstrated the benefits of using an IWB, such as assisting with whole class teaching, differentiating instruction, and integrating texts, images, and videos (Wall, Higgins, & Smith, 2005; Wood & Ashfield, 2008). IWBs provide visual assistance for learners because of the way the information is displayed and how students can physically interact with the board (Hur & Suh, 2012; Wood & Ashfield, 2008). Displaying images allow ELL students to link their native language and English, which can help with vocabulary skills (Hur & Suh, 2012). Digital technology such as IWBs provided teachers with the ability to incorporate audio and visual data into their instruction, which can be beneficial to improving English vocabulary development (Hur & Suh, 2012). The IWB can help students to develop their vocabulary through the combined use of audio, visual images, tactile use, and teacher-assisted instruction.

Definition of Terms

Abu Dhabi Education Council (ADEC). The head government agency that oversees education in the Emirate of Abu Dhabi (ADEC, 2013). ADEC provides policy for K-12, colleges, and universities.

Basic Interpersonal Communication Skills (BICS). BICS refers to language that is used in a social or conversational setting (Cummins, 2002). This is the very basic level of vocabulary that students are required to know in English.

Cognitive Academic Language Proficiency (CALP). CALP refers to complex language that is used in academic classes such as mathematics and science (Cummins, 2002).

English Language Learner (ELL). This is a person who is learning English in addition to their native language (Krashen, 1983).

External Measurement of Student Assessment (EMSA). These are a battery of tests that students take in English, Math, Science, and Arabic every year in Abu Dhabi (ADEC, 2013).

Interactive Whiteboard (IWB). An IWB is a large display that is connected to a computer that allows a user to control the computer using a pen, finger, or other device (Wall, Higgins, & Smith, 2005).

Purpose of the Study

Assisting students with their vocabulary development is one of the goals of the district in which I worked. The purpose of this study was to investigate how the use of IWB materials can assist with vocabulary development for ELL students in a boys school in the United Arab Emirates.

Research Question

This study investigated the following question:

How can IWBs be used to facilitate vocabulary development in an ELL classroom?

For my grade level, I was responsible for designing and planning all of the literacy-based lessons. Therefore, this study focused on vocabulary instruction using available resources in each of the classes (IWBs). The teachers in this study were responsible for planning mathematics and science lessons.

IWB Software and Materials

Teachers can develop students' language skills in writing, reading, listening, and speaking by making language comprehensible through modification of input and the use of nonverbal aids such as visuals and graphic organizers. The first lesson focused on visual scaffolding, which involves images and words that are seen and heard. This was one way to provide comprehensible input to ELL students so that they progress in their English language development (Short, Echevarria, & Richards-Tutor, 2011). The second lesson focused on the tactile aspect of the IWB, which gave students an opportunity to practice their vocabulary. The IWB assisted with vocabulary development because of the emphasis on the use of visuals, realia, and inclusion of high-frequency vocabulary words.

Methodology

This was an exploratory study that used qualitative methods to research the facilitation of English vocabulary development using an IWB. Participants included two native English-speaking teachers who each taught two classes of approximately 25 grade three ELL students (four total classes) at Sas Al Nakhl School in Abu Dhabi.

Data Collection

Data for this study were obtained from four sources (Glesne, 2006), including teacher interviews, classroom observations, a reflective journal, and a recording of the IWB lesson.

- Teacher Interviews: Participating teachers were interviewed three times during this research project through the use of a semi-structured, open-ended interview protocol (Appendix A). These interviews provided insight into the teachers' perspective on usage of an IWB for vocabulary development.

- Classroom Observations The lessons were observed using an observation tool, which is included in Appendix B. The tool combined aspects from the Sheltered Instruction Observation Protocol (SIOP) model and the International Society for Technology in Education Classroom Observation Tool (ICOT) (Echevarria et al., 2000; ISTE, 2013). It featured sections on feedback on IWB instruction for whole group/mini-lesson, small group instruction, closing, scaffolding, interaction with language and content, application, and practice (Echevarria, Short, & Vogt, 2008; Fledderus, Knoppert, & Soesbergen, 2011; ISTE, 2013). This observation tool was used to observe the lesson. Classroom observations were also used to guide the second interviews with teachers and to inquire about the activities and lesson implementation.

- Recording of Lesson: The student and teacher interaction with the board were recorded using visual capture software, and the recording focused on the different aspects of the lesson.

- Reflective Journal: A journal was used to reflect on the observation of the IWB activities, classroom, and teacher interviews. It helped me keep notes of my conversations with teachers during the planning phase, my reflections on classroom observations and any notes I made during interviews. I was embedded in the context as a teacher who planned the lessons and taught the same grade level. I was also the researcher. A journal assisted in identifying my bias or attitudes that could influence the data analysis and helped me conduct effective qualitative research (Glesne, 2006; Saldana, 2013). Emerging patterns and ideas were included in this journal.

Data Analysis

Because this dissertation was an exploratory, qualitative research project, teacher interviews, classroom observations, and a reflective journal were analyzed according to qualitative research analysis procedures (Glesne, 2006; Patton, 1987;

Saldana, 2013). Initially the data from these sources were analyzed together to look for patterns. The goal was to find patterns of action in the teachers' and students' behavior and interaction with the IWB in the data. This dissertation utilized thematic analysis for analyzing the qualitative data (Braun & Clarke, 2006). Mertler (2006) defined coding as a system of categorization that is used to group data in order to find similar types of information. Connections were made from the data, and each of these components provided evidence on how IWBs can be used to facilitate vocabulary development.

Trustworthiness of the data was achieved through persistent observations, triangulation of the data sources, clarification of researcher bias, and rich descriptions of the subjects (Glesne, 2006). The findings from this study were transferable because the findings of this study can be applied to other classrooms because of the rich description of data and context. The credibility of the study was increased because I shared and discussed the results of the observations with the teachers. The limitations of the study included focusing solely on the usage of the IWB for vocabulary development where there could be other variables that affect student vocabulary development in the classroom. Another limitation was the small sample size of two teachers and four classrooms, as well as researcher bias.

Significance of the Study

The purpose of this exploratory, qualitative study was to determine best practices for designing lessons and activities for vocabulary development of ELL students utilizing the IWB. In the United Arab Emirates, there were IWBs installed in all of the public schools, but there was no research on best practices for using these pieces of technology even though one of the goals of the Abu Dhabi Education Council was to use this technology in order to assist students with vocabulary development. This study

adds to the research about the use of IWBs for ELL students. This study also adds to research about comprehensible input and the use of nonverbal aids such as visuals and graphic organizers. ELL students in various contexts will be assisted by this research.

CHAPTER 2
LITERATURE REVIEW

Using second language acquisition theories and prior research on IWBs, this literature review synthesized how IWBs could be used to assist students with vocabulary development. The connections between vocabulary development and comprehensible input were discussed, as well as strategies for working with English as Language Learners. Studies have shown that usage of IWBs by ELL students can assist teachers with whole-class and small-group instruction, reach multiple learning styles and modalities, and support the student learning process through authentic contextualized learning (Ansell & Park, 2003; Beeland, 2002; Levy, 2002; Reaume, 2007; Schmid, 2010; Williams & Chang, 2009). Second language acquisition theorists such as Krashen and Terrell (1983) and Cummins (2002) have made reference to different methods and styles of learning for second language acquisition. Making language comprehensible for learners and providing explicit vocabulary instruction were also discussed (Beck, McKeown, & Kucan, 2002).

This chapter contains five sections: Education in the United Arab Emirates, Second Language Acquisition, Theoretical Framework, Literature on the IWB, and the Significance of Studying the IWB.

Education in the United Arab Emirates

Education in the UAE began in a splintered fashion as the United Arab Emirates, before this was known as the Trucial States, became a country in 1971 (Ridge, 2009). Historically, the Arabian Peninsula was split according to tribes. Education in the format of traditional schools did not come until 1953 when schools were built. Students were not required to attend, and usually only wealthier citizens attended. In 2005, the Abu

Dhabi Education Council (ADEC) was formed to develop the educational system in Abu Dhabi. Within the last five years, ADEC hired over 2,000 native English speakers to teach English, mathematics and science in schools. Abu Dhabi is unique in that 90 percent of the Emirate's workforce consists of foreign workers (Boyle, 2011). Although Arabic is the mandated language of government, English is the lingua franca in the majority of communities in Abu Dhabi and the United Arab Emirates. Industries such as oil, gas, banking, IT, air transport, and higher education all use English as their method of communication.

In second language acquisition literature, there was a difference between the use of the terms English Language Learners (ELL), English as Foreign Language (EFL) and English as a Second Language (ESL). ESL refers to learning a language in an area where both languages are used in the context of everyday life. For instance, in many parts of the southwestern United States, the population speaks both English and Spanish. EFL refers to learning a language in a country but not necessarily needing to use this language in the context of everyday life. For instance, a student living in England and studying Farsi, which is not needed to communicate.

Gardner (2001) expressed concern with the differences between EFL and ESL when he made the case that the social context can well influence the dynamics of learning a second language. He included an example of Canada being a bilingual country in which two languages, English and French, are spoken but are not necessarily available in every individual's environment in the country. The same was true in Abu Dhabi. Students were exposed to English in various contexts every day. However, for the purpose of this study the students will be referred to as ELL students since the

context is very linguistically complicated. ELL is more commonly used in K-12 learning environments and can encompass some of the features of ESL and EFL. Some of the students were able to speak English on a daily basis at home whereas many were not able to.

One of the issues of contention in the Middle East was the difference in education outcomes for boys and girls. According to the Arab Human Development Report (ADHR), boys in the Middle East performed at a lower level than girls (UNDP, 2006). In many nations in the Middle East, such as the UAE, Bahrain, and Kuwait, girls attended university at a greater rate than boys (Ridge, 2009). Research on education in the Middle East has tended to focus on test scores, and while there was research on the use of technology in the classroom (Cloke & Al-Ameri, 2000), a larger issue was the lack of research on increasing performance with learning outcomes for boys. In that context, this research took place in a boys' school in the UAE.

Educational Technology in the Middle East

In many Middle Eastern countries, the use of educational technology fell behind other countries. So many children learn more about how to use technology outside of the classroom (UNESCO, 2013). Children who were economically struggling likely will not have access to technology in the Middle East. Countries such as Jordan, Qatar and Oman have included recommendations in their national curriculums to use technology in instruction (McKinsey & Company, 2005).

The Kingdom of Jordan has a framework entitled the Jordan Education Initiative (JEI), which focused on integrating technology into the curriculum (UNESCO, 2011). The JEI piloted this program in 100 schools around the country and also put out a Curriculum and Learning Assessment Framework that helped with designing lessons

around technology. This was in contrast to countries such as Egypt and Palestine where the use of technology in instruction was varied across subject areas.

An issue in many of these countries was the availability of infrastructure to support the learning process. Putting technology into schools requires a mixture of stable electricity sources, which many communities do not have (UNESCO, 2013). The resources that are required to sustain a technology program in schools are sizable. In the integration of technology for teaching curriculum, the United Arab Emirates and, within it, Abu Dhabi are ahead of other Middle Eastern countries (Abu Dhabi Education Council, 2013).

Educational Technology in the United Arab Emirates

In the United Arab Emirates and the Emirate of Abu Dhabi, ADEC has undertaken a major initiative to use technology to promote teaching and learning in schools (Al Mekhlafi, 2006b). Several professional development companies and universities have received contracts to deliver training to teachers in Abu Dhabi schools (Abu Dhabi Education Council, 2013). Not only does the training focus on ELL specific strategies, but it is also inclusive of educational technology based professional development. There are also several workshops and conferences that deliver training on different topics in the UAE.

In the UAE, several researchers have conducted studies about using Educational Technology at the elementary, secondary, and university levels (Al Mekhlafi, 2006a; 2006b; Al Ghazo, 2006a; Al Ghazo, 2006b). Two studies that researched Computer Assisted Language Learning (CALL) and ESL learners in the UAE found that students performed better when technology was used (Al Mekhlafi, 2006a; Al Mekhlafi, 2006b). Several other studies focused on teacher beliefs and found that there was a direct

correlation between buy in and effective usage of the technology (Al Ghazo, 2006a; Al Ghazo, 2006b; Ismail, Al Mekhlafi, and Al Mekhlafi, 2010; Serhan, 2007).

In order for Educational Technology to be effective in classrooms, both school leaders and teachers have to buy in to the usage of technology for students. Serhan (2007) conducted a study about school principals in the UAE and their attitudes toward the use of technology in schools. The study measured the effectiveness of a technology training workshop. A questionnaire was designed and given to 200 school principals in the UAE. The study showed that principals were willing to use technology in teaching and that they supported using technology in the school. In another 2010 study, Ismail, Al Mekhlafi, and Al Mekhlafi studied perceptions of Arabic and English teachers about using technology in classes in the UAE. In this mixed methods study, a questionnaire and a focus group interview were utilized and analyzed. Several themes and results emerged from their data, and they found that not only were beliefs important but so were barriers to technology use and having incentives for teachers to use technology. ADEC's educational policy for teaching reading and writing in a second language has been greatly enhanced by the implementation of technology in the classroom.

In another study, Al Ghazo (2006a) researched quality of Internet use by teachers in the UAE. In this quantitative study, Al Ghazo examined several of the obstacles that teachers faced when using the Internet. He collected data from over 400 elementary and secondary teachers using an online survey. The study found that Internet use was very minimal in UAE classrooms and that there were several obstacles to use, such as limited access, lack of expertise, and lack of time. In another study, Al Ghazo (2006b) looked at student attitudes toward web-based instruction at the

university level for student teachers. The sample consisted of over 60 college female students and used a survey with five point Likert items as well as open-ended questions to obtain data. The study presented several positive aspects of using educational technology in classrooms, such as online discussion boards, access to course materials, and communication and collaboration with classmates.

Educational Technology and ESL in the United Arab Emirates

The most prominent researcher on using technology in schools for English language teaching in the UAE is Abdurrahman Al Mekhlafi, a professor at the United Arab Emirates University in Al-Ain. In a quantitative study, Al Mekhlafi (2006a) looked at the effect of Computer Assisted Language Learning (CALL) on elementary-prep school students' improvement in English. On conducting an analysis of variance (ANOVA), he found there was a large difference between CALL users and nonusers, with the experimental group performing better. The students were administered a questionnaire where the results showed that the experimental group had a more positive attitude towards using the technology.

In a similar study, Al Mekhlafi (2006b) investigated the effect of an interactive multimedia program (IMM) on student's acquisition of ESL skills in the UAE. Although this group is older, it still has relevance to the students at my school (Al Mekhlafi, 2006b). In this study, 90 sixth-grade students were split into experimental (n=46) and control groups (n=44) (Al Mekhlafi, 2006b). The study found that there was no significant difference between IMM users and nonusers in terms of ESL skills; however, students who displayed a greater cognitive propensity toward animation performed at a greater level (Al Mekhlafi, 2006b).

In a mixed methods study, Al Mekhlafi and Al Meqdadi (2010) investigated the technology integration at UAE model schools, which are similar to my school environment. This study researched 40 female and 60 male teachers using focus group interviews and questionnaires. The study showed that teachers at both of the schools are integrating technology in their class activities; however; male teachers and female teachers used different instructional methods. A major finding is that the technology must be matched with the curriculum goals so that technology integration can enhance teaching and learning (Al Mekhlafi & Al Meqdadi, 2010). These three studies were directly related to the type of school in which I worked, and they provide a basis for different methods to use technology at this level (Al Mekhlafi, 2006, 2006a; Al Mekhlafi & Al Meqdadi, 2010).

When looking at the studies by Keblawi (2005) and Qashoa (2010) as well as observations from my classroom, I focused on designing the lessons and activities around vocabulary because this is one of the areas where many of my students struggle. Although these studies do not revolve around the IWB, they were still relevant to understanding ways in which technology has been integrated into classrooms in the UAE.

IWBs in the United Arab Emirates

While there is research on IWBs in the United Arab Emirates, much of it was done at the university level. Al Qirim (2011) studied IWBs at the university level and found that students were successful when using it. In an additional study on research and motivation, Hamdan, Al-Qirim, and Asmar (2012) looked at different ways that the IWB impacted general education students. Future educators in the UAE and their beliefs about using IWBs were also studied (Al Ghazo, 2006b; Makrakis, 2005).

Educational technology research based around the IWB in the UAE has focused on teacher education and research at the university level (Al Ghazo, 2006b; Al Qirim, 2011; Hamdan, Al Qirim, & Asmar, 2012). My research focused on the use of the IWB at the elementary level. Much of the research that was presented in these studies focused on the areas that influenced my research: new teaching pedagogy such as scaffolding, learning by doing, and hypermedia presentation.

Second Language Acquisition

This dissertation utilized research by Stephen Krashen's (1981) and his work on language acquisition and the monitor model, which consists of utilizing input rich language in order to help a student become fluent in the language without direct teacher intervention. Krashen's theory includes five hypotheses (Krashen, 1981; Krashen & Terrell, 1983).

Krashen's Hypotheses

The first hypothesis is the acquisition learning hypothesis, which is that language acquisition is the result of a progression in the sub consciousness, which is comparable to what happens when children first learn their first language. This hypothesis requires a form of natural communication. This concept of language learning acquisition is generally accepted by second language practitioners and researchers around the world (Krashen & Terrell, 1983). The second hypothesis is the monitor hypothesis, which is the link between learning and acquisition (Krashen & Terrell, 1983). Within this hypothesis, there is a monitor (teacher) who is used as a device that learners use to edit their language performance (Ellis, 1998; Krashen & Terrell, 1983).

Krashen and Terrell's (1983) third hypothesis is the natural order hypothesis, which is that grammar acquisition follows a predictable order for different languages.

Certain grammatical structures can be attained in a predictable order. The speaker thus acquires language through speaking in everyday situations (Krashen & Terrell, 1983). The fourth hypothesis is the input hypothesis (Krashen & Terrell, 1983). According to the input hypothesis, second language acquisition is the direct outcome of a learner's understanding of the target language within natural communication settings (Krashen & Terrell, 1983).

The fifth hypothesis and perhaps the most important is the affective filter hypothesis (Krashen & Terrell, 1983; Valdez, 2012). According to the affective filter hypothesis, how an individual feels about the second language can affect his or her learning of the second language (Krashen & Terrell, 1983). In this hypothesis, Krashen argues that learning a new language is not like other subject material because one must publicly practice his or her knowledge. A second language can best be acquired when the conditions in the environment are comparable to first language acquisition (Krashen & Terrell, 1983). According to the affective filter hypothesis, instruction should be focused on meaning rather than grammar (Krashen & Terrell, 1983; Valdez, 2012). It is important for students to be comfortable in their settings in order to reduce anxiety and nervousness, which can hinder acquisition (Krashen & Terrell, 1983; Valdez, 2012).

Vocabulary as a Key Part of Second Language Acquisition

Second language acquisition theorists generally agree on the following accepted stages of development through which second language learners' progress: Preproduction; Early Production; Speech Emergent; and Intermediate (Diaz-Rico & Weed, 2010; Krashen & Terrell, 1983). The Preproduction stage is "called the silent period when the learner is absorbing the sounds and rhythms of the new language, becoming attuned to the flow of the speech stream, and beginning to isolate specific

words." During this stage, the learner relies on contextual clues for understanding key words and generally communicates nonverbally (Diaz-Rico & Weed, 2010, p. 9).

The Early Production stage is when a learner feels more confident in using words and phrases. Learners use two- and three-word sentences during this phase. During the Speech Emergent stage, learners respond using more complex language. The Intermediate Stage students can engage in conversations and correct their own mistakes and errors (Diaz-Rico & Weed, 2010). Second language acquisition is a long process that is very complex, and it is necessary to assist students with vocabulary development through implicit and explicit instruction (Asbeck, 2012). This dissertation focused on the following ways that the IWB can assist with development of vocabulary: learning the meaning of new words, associating words with images, and the presentation of word families.

Learning the Meaning of New Words

Vocabulary development and acquisition has been indicated by Stahl and Fairbanks (1986) as the most important aspect of acquiring proficiency in spoken language. Based on research conducted by Nagy and Anderson (1984) as well as Stahl and Nagy (2006), children must learn over 180,000 different words in order to be successful. Although direct vocabulary instruction cannot cover the majority of these words, it is still important to implement activities that assist students with developing their vocabulary.

Beck et al. (2002) has put forth that teachers should select words for instruction that students will need to comprehend and understand text. Words are separated into three tiers. The first tier focuses on basic words such as boy, girl and baby. Tier two focuses on words that are more complicated such as strange and privileged. Tier three

words are specific to certain academic contexts (Beck et al., 2002). This study has a focus on tier one words since these are words that the students need to have a base understanding of in order to be successful as ESL students.

Associating Words with Images

Second language learners that are beginners benefit tremendously from learning basic vocabulary in a contextualized and decontextualized manner. In a 1999 study, Chun and Plass found that students benefited from having the picture with the first language translation included produced great gains in their assessment outcomes. However, students that might not necessarily have a large vocabulary can benefit from learning vocabulary in a decontextualized setting (Kopstein & Roshall, 1954; Wimer & Lambert, 1959).

The IWB incorporates using visual techniques when teaching vocabulary. This can include photos, drawings, and flashcards (Interactive Technologies in Language Teaching, 2011). The ability to move objects on the screen with the drag and drop tool is very helping when matching words with the pictures. The IWB is very helpful when supporting vocabulary learning with visual input.

Presentation of Word Families

The text highlighting feature of the IWB software can assist with learning the different rhyming formations of word families. Word families are groups of words that rhyme and sound similar; the difference being the first consonant or blend at the beginning of the word (Scheiner, 2005). An example would be the word family “-at,” which consists of the words “bat,” “cat,” “fat,” “hat,” “mat,” “pat,” “rat,” “sat,” and “vat.” Teaching word families can help with phonemic awareness with English language learners (Edelen-Smith, 1997). IWBs can assist with this aspect of vocabulary

development by having the teacher/students highlight the word families as well as match them (Interactive Technologies in Language Teaching, 2011).

Dimensions of Language Proficiency

Cummins (2002) identified two concepts for language proficiency in Basic Interpersonal Communicative Skills (BICS) and Cognitive Academic Language Proficiency (CALP). BICS refers to language that is used in a social or conversational setting. This type of language is acquired easily through the use and acquisition of simple sentence structures and high frequency words (Cummins, 1981). English language learners take one to two years of second language exposure in order to attain this level. CALP requires students to understand vocabulary that is not used as frequently for complex language (Cummins, 2002). Academic language that is used in classes such as mathematics and science require the usage of this type of vocabulary. Acquiring and using academic language is a complex process, and English language learners may require three to five years to reach proficiency (Smiley & Salsberry, 2007).

Theoretical Framework

The theoretical framework used in this study consists of Comprehensible Input and Explicit Vocabulary Instruction to inform IWB lesson design. The previous section focused on research about educational technology in the Middle East and the UAE. Second language acquisition principles were also included. This section described the components that informed the IWB lesson design of the study.

Comprehensible Input

One of the most influential hypothesis on second language acquisition is comprehensible input, which was developed by Krashen and Long (Ellis, 1998; Wu, 2010). The Input Hypothesis puts forward that “learners progress along the natural

order by understanding input that contains structures a little bit beyond their current level of competence" (Ellis, 1998; Wu, 2010). One of the most important approaches for a learner to acquire language is to understand the level of input. When the input is internalized, then language acquisition can occur. Krashen and Long put forth that comprehensible input refers to language materials that are beyond the learner's current level (Ellis, 1998). However, if it is too advanced for the learner, then it has no significance for language acquisition (Wu, 2010).

Barton and Griffin (2009) conducted a study involving second language learners and found that it is important to give students support to read, write, verbalize, and listen in the classroom. Effective teachers emphasize problem solving with students by engaging in instructional conversation as well as reading and writing across the curriculum. Echevarria, Short, and Vogt (2008) put forward several instructional techniques that help students to understand and comprehend instruction. These techniques included (a) speech appropriate to the student's English proficiency level, (b) academic tasks that are concise, (c) modeling concepts, (d) the use of visuals, body language and hands on activities, and (e) giving opportunities for discussion and concept clarification (Vidot, 2011).

These following two studies were relevant because of the importance and role of input in second language acquisition. Two areas of study were quantity and quality in reference to input. Bialystok (1978) and Seliger (1977) have put forth evidence that learners who have the opportunity to engage in high levels of target language input will have greater proficiency in that language. Second language learners who were exposed to more input from inside and outside the classroom environment displayed a greater

proficiency than those who were not as exposed. However, research by Snow, Van Eeden, and Muysken (1981) indicated that the quantity of second language input in the target language is not necessarily the most important aspect because the quality of the input was just as important.

Larsen-Freeman (1976) researched input in the ESL classroom and found that frequency of correct input was the main determinant of how well the learner produced language. Although frequency of exposure may create a more accurate production of language, a causal but not necessarily correlational relationship between quantity and quality of input still exists. Krashen (1981) hypothesized that the input must be comprehensible to the learner, and this may be achievable through modifying input. The IWB can assist in this area because of the ability of the learner to make direct changes immediately.

Explicit Vocabulary Instruction

Beck et al. (2002) provides a definition of explicit vocabulary instruction that defines the instruction as giving rich information about words and their uses, giving students opportunities to think about and use words. Explicit vocabulary instruction also gives students the opportunity to enhance their comprehension and production. One of the major features of explicit instruction is using the targeted words in a way that is decontextualized. The explicit method of teaching vocabulary is considered effective because it stresses exposure to targeted vocabulary, words in contexts that are meaningful, and engaging students in the learning process (Longo & Curtis, 2008).

Explicit vocabulary instruction is explicit and direct. The teacher explains the word to the students and gives them an opportunity to make applications of their knowledge of the word. Students should be given multiple opportunities to practice with

the word in order to remember it (Graves, 2006). Research has shown that exposing students to words in different contexts allows the students to use their background knowledge with the word (Beck et al., 2002). Students should also be given pictures of target words because pictures contain a rich source of semantic information (McGregor, Sheng, & Ball, 2007).

Literature on the IWB

IWBs can help to enhance the interactive nature of classrooms in the 21st century. By using the IWB, teachers and students highlight, edit, modify, write notes, add sound, and add video by using the board (Ansell & Park, 2003). The IWB has three different parts: a data projector, a computer, and the IWB. Using the IWB consists of touching the board by utilizing either an interactive marker or with one's hand to interact with the computer in a similar way to a mouse. IWBs provide the opportunity to increase levels of attention and assist students with their studies (Gerald, Greene & Widener, 1999; Knowlton, 2008; Tanner & Jones, 2007).

IWBs also have several benefits for teaching through whole class instruction. Wall et al. (2005) looked at the potentially positive benefits of using the IWB through flexibility and versatility, multimedia/multimodal presentations, efficiency, supporting planning and the development of resources, modeling ICT skills, and interactivity and participation in lessons. There are four main aspects of whole-class teaching that practitioners can use the IWB to demonstrate: to use software, to present and discuss information, to facilitate the explanation of ideas, and to enable students to show and receive feedback on their work (Levy, 2002).

The positive and negative aspects of IWBs in primary schools was researched by Fekonja-Peklaj and Marjanovic-Umek (2015) in a qualitative study. Data were collected

from the teachers and students through observation and interviews. Positive aspects were found to include benefits for motivation and engagement and displaying content. Several negative aspects were difficulties with technical problems and too much reliance on whole class teaching. In another study, Hennessy (2011) examined how an IWB could be used to assist student learning through classroom dialogue. By analyzing lessons, the study showed that teachers could use the IWB by capitalizing on learners' thinking as it emerges rather than focusing on finished work products. The study analyzed different case studies from classrooms in the United Kingdom and showed that as lessons progressed over time, the ability of the IWB to manipulate digital artifacts opened up dialogue in the classroom.

Stout (2013) conducted a quantitative study that examined teaching with the IWB and its impact on student outcomes for economically disadvantaged students. The study conducted a comparison group study of two fifth grade classes using the student standardized state assessment data. Using independent sample T-tests, the study revealed that students that were taught with IWBs earned higher scores on the standardized tests than students that were taught without IWBs. James (2013) also found that teacher beliefs about using IWBs as a resource had a direct impact on raising student achievement levels. He conducted research on teacher beliefs and challenges of using IWBs by observing ten teachers and having them participate in open-ended interviews. A mixed methods approach was used by Stewart (2013), who studied 50 teachers to identify how well IWBs were integrated in a school, as well as their impact on student engagement. Each of the teachers was observed twice for 30-

minute intervals. The study found that there was a direct relationship between training, daily use of the IWB and their beliefs about technology use in the classroom.

Teachers also viewed IWBs as enrichment in class in Hilbert and Gunther's (2014) empirical study on IWBs in Germany. The 158 teachers that were surveyed stated that even though they had a higher level of IWB competence, they did not necessarily make application of their skills in their daily classroom routine. A study by Sweeney (2008) created a framework for teachers to effectively utilize the IWB in their practice utilizing empirical data over 18 months from an Australian elementary school. Sweeney (2008) argued for the creation of a developmental framework for using IWBs for teachers. Additionally, the importance of professional development for teachers emerged in a study about using IWBs for English teaching and language. Yang and Teng (2014) conducted a qualitative study where they interviewed six English teachers for 30 minutes each in order to gain an understanding of their opinions and attitudes toward using IWBs. The results of the study showed that professional development is important to effectively using the IWB because a teacher's mastery of IWB skills and knowledge of teaching determined how well the technology was used in the classroom. Yang and Teng (2014) also found that collaboration and interaction between teachers and students increased as IWBs were used more frequently.

IWBs were also found by Warwick, Mercer, Kershner, and Kleine Staarman (2010) to be useful for establishing collaborative classrooms. They conducted a study on using the IWB to establish teacher and student collaboration. This mixed methods study utilized analysis of video and other types of data from science classrooms in the United Kingdom for grade four and five students. The study found that having the

teacher guide the student interactions at the board was effective in getting the students to collaborate at a high level.

IWBs and Learning Styles

The IWB can be used to teach students in many different ways, which are inclusive of the three modalities of learning (Cuban, 2001). Visual learning is the first modality, which uses pictures, text, video, and animation. Auditory learning is the second modality, and activities that include auditory learning use words for speeches and pronunciation. The third modality is learning through touch or tactile. Students using an IWB can physically interact with the board, which can help meet with the learning needs of tactile learners. There are many software programs available that allow students to interact with the IWB in a meaningful way.

Since the 1920s, research about learning styles, such as visual, auditory and kinesthetic, asserts that every individual has a particular style that is well suited for comprehending and retaining information (Chapman, 2005). Therefore, teachers should strive to develop instructional practices that keep these concepts in mind. Most students use different parts of each learning style, and usually one or two of these styles are dominant (Glover & Miller, 2002). The IWB can assist with all three of these styles. Students who learn best through touch can use the board to make marks and to interact with different activities (Bell, 2002). Students who are more auditory can learn from the discussions and conversations that come up when using the IWB, and students who are more visual can see what is happening during the lesson. The IWB is one of many different resources that teachers can use to assist with the different learning styles of students. Effective usage of the IWB depends on being informed about principles of instructional design and implementation.

IWBs and Academic Subjects

Teaching academic subjects through using the IWB can help with teaching academic subjects (Thompson & Flecknoe, 2003). In a 2007 study, Glover, Miller, Averis and Door found that the usage of technology allowed teachers the ability to integrate resources into their lessons. These teachers stated that through the usage of the IWB, they were able to reach many different styles of learners, and using the IWB's software affords teachers the opportunity to reach each of their students' learning styles (Glover and Miller, 2002). Using the IWB during instruction also leads to greater student participation (Jones, 2004). The British Educational Communications and Technology Agency (BECTA) in 2003 conducted a survey of primary schools and found some good practices from using the IWB. Students were more engaged in the lessons with the IWB rather than with traditional classroom teaching (Jones, 2004). From analysis of their study lessons, Miller et al. (2004) found several takeaways from the usage of IWBs. Multiple learning styles, engaging students, giving feedback, understanding of metacognition, and creating sequential lessons were all important to the effectiveness of using the IWB (Miller et al., 2004).

Goodwin (2008) researched the effects of multimedia tools on kindergarten students' representations of fractions. The study compared two classes in which one class received whole-class technology based mathematics and the comparison class kept their regular mathematics program. Goodwin found that students that received instruction with IWBs and other interactive tools had greater growth in their cognitive and symbolic abilities than those in the comparison group. Preschool teachers using IWBs in mathematics education were also studied by Bourbour, Vigmo, and Samuelsson (2015), whose qualitative data indicated that the use of IWBs could be

categorized into three areas. Students were engaged through multisensory resources, engaging in problem solving activities, and finding student interest. Data were collected from observation of three preschool teachers in Sweden. The study supported the importance of teacher's background knowledge about using IWBs.

In a 2015 quantitative study, Erbas, Ince, and Kaya explored the effect of using an IWB in contrast to a direct instruction environment in a high school mathematics course in Turkey. The study looked at student attitudes toward using IWBs as well as their performance on pre- and post- tests. The study found that student performance increased from pre- test to post- test and also displayed greater retention in the IWB group. The student's also had a greater attitude toward using the technology in their classroom. The effects of an IWB on student achievement were also studied by Marzano and Haystead (2009) who examined 50 schools in the United States in an evaluation study that had 1,622 students in the control group and 1,716 students in a control group. The treatment group used the IWB, whereas the other group used its regular mathematics materials. They found that there were several factors that related to increased student achievement and math with the IWB being that (a) a teacher has more than two years of experience using the technology and has high confidence in his or her ability to use it and (b) the teacher uses the technology more than 70 percent of the time in the classroom.

IWBs and English as a Second Language

Recent research suggests that technology usage in foreign language or ELL classrooms should be centered around the learner, be authentic, and be tied to making a difference in the student's place in the world (Schmid, 2010). This is the socio-cognitive approach wherein the classroom is viewed as a place of interaction in which

the learner is constantly negotiating meaning within contexts that are authentic. Schmid (2010) focuses on the design, management, and balance of the use of technology. Traditionally the IWB has not been used in this format and has primarily been used in a behavioristic manner, which focuses on control of behavior and linguistics (Warschauer, 2000). Teachers have used IWB technology in order to support the learning process of students through presentations and materials that were carefully planned (Gray, Hagger-Vaughan, Pilkington, & Tomkins, 2007).

Gray et al. (2005) found that the majority of teachers in a British high school carefully planned lessons around the IWB and that the whiteboard was mainly used for practice activities. Schmid (2010) argued that teachers must avoid excessive use of pre-prepared resources in order to prevent cognitive overload for students. She also found that there are different opportunities for using IWBs for constructivist practice and also to help with learners' active participation in the classroom. This research also found that teachers could use the technology to attain flexibility in their lessons through the use of Web 2.0 resources. The teachers used the technology to go beyond using the IWB as a tool for presentations (Williams & Chang, 2009). Schmid (2010) and Williams and Chang (2009) have put forth the argument that high quality professional development is needed for teachers to integrate IWBs in a socio-cognitive learning environment.

In a 2010 study, the Round Rock Independent School (ISD) conducted a study on using IWB technology for second language learners (Lopez, 2010). The goal was to reduce the parity between native English-speaking students and second-language learners. The study used a quasi-experimental design with students in grades three and five, which compared ELL students in traditional classrooms to those in classrooms

equipped with IWBs. Lopez (2010) found that IWBs can assist with performance. The study conducted an evaluation that strongly argued that IWBs could close the achievement gap between regular and ESL student. The literature points out that IWBs assist in a positive manner with whole-class language classrooms because of the integration of digital media, websites, and the ability to manipulate vocabulary (Hall & Higgins, 2005; Schmid, 2010; Walker, 2003). Vocabulary learning is enhanced when words include visual and verbal coding as IWBs can assist in this area (Plass, Chun, Mayer & Leutner, 2003; Walker, 2003).

Potential Limitations of IWBs

One of the major issues raised by teachers related to the use of IWBs is the need for training (Wall et al., 2005). Teachers initially show enthusiasm but then trail off as the support decreases. Levy (2002) observed that teachers who were confident in their usage of ICT were able to develop their own uses of the IWB following initial training. These are some of the areas where training and professional development were an issue. Another common problem cited in the literature is related to the placement of IWB equipment in the classrooms (Wall et al., 2005). Visual problems arise when the sun shines directly on the board, and there is also concern for the health and safety of the students due to the number of wires that are required for the IWB (Levy, 2002). The height and placement of the IWB can also be an issue particularly if the students cannot reach the top of the whiteboard or if it is too low, limiting the view for students sitting in the back (Wall et al., 2005).

Opportunities for learning invariably depend on the use of the IWB by the teacher. IWBs offer many different possibilities that require specific knowledge on how to use their versatility in the classroom. Teachers in second-language and immersion

classrooms must have knowledge of the relationship between processes and children's learning that can help them to create opportunities for verbal interaction, which can complement and build on a child's tactile interaction with the IWB (Coyle, Yanez, & Verdu, 2010).

Significance of Studying IWBs

In Abu Dhabi, IWBs were installed in every classroom in 2010. Literature has shown that the IWB can be very beneficial for enhancing the interactivity of classrooms (Ansell & Park, 2003; Gerald, Greene, & Widener, 1999; Tanner & Jones, 2007). Research on IWBs and ESL learners has shown that students can be successfully supported through the learning process (Gray et al., 2005; Lopez, 2010; Plass et al., 2003; Schmid, 2010).

Although there is much research on the usage of computers in classrooms in the UAE and Middle East, the research was lacking in the areas of using the IWB to teach vocabulary to second-language learners in these types of settings (Al Mekhlafi, 2006; Al Mekhalfi & Al Meqdadi, 2010; Qashoa, 2010). Much of the research on IWBs in the UAE has been conducted about teacher beliefs and with university general education students (Al Ghazo, 2006b; Al Qirim, 2011; Hamdan, Al Qirim, & Asmar, 2012). In general educational technology, Keblawi (2005) and Qashoa (2010) both have conducted studies around motivation of UAE students, which were helpful in my thinking about designing lessons in this study.

Classes in the UAE have 25 to 35 Arabic-speaking ELL students who are taught by one teacher, whereas most studies on explicit vocabulary instruction, IWBs, and second-language acquisition and development focused on teaching ESL in a mainstream setting (Beck et al., 2002; Glover & Miller, 2002; Levy, 2002; Longo &

Curtis, 2008). Research on the potential and limitations of IWB use for vocabulary development for ESL learners had mainly been conducted outside of the UAE, which has a unique context (Gray et al., 2007; Hall & Higgins, 2005; Lopez, 2010; Plass et al., 2003; Schmid, 2010; Walker, 2003; Warschauer, 2000; Williams & Chang, 2009). This study contributes to the literature that is available both about educational technology and IWBs for ELL students in the UAE. The following chapter explains the design of IWB activities based on the literature reviewed, followed by the research design used in this study.

CHAPTER 3
METHODOLOGY

This study utilized an exploratory, qualitative research approach to examine how the use of IWBs can assist with vocabulary development for ELL third grade boys in a school in the United Arab Emirates. As a first step to achieve this goal, two lessons were created for the IWB based on research on comprehensible input, explicit vocabulary instruction, and literature on the IWB (Beck et al., 2002; Echevarria, Short, & Vogt, 2008; Interactive Technologies in Language Teaching, 2011). The research question for this study was:

How can IWBs be used to facilitate vocabulary development in an ELL classroom?

This research question is explored in an open-ended manner through the collection of qualitative data from teachers and third grade classrooms that utilized IWBs to teach ELL students In qualitative research, “engagement and extensive observation are central,” and ethnographic methods such as interviewing “can be used to help us understand the social interactions in the environment” (Fasse & Kolodner, 2000, p. 193).

Chapter 3 first includes a brief overview of my background as the researcher and is then divided into two sections. The first section describes the design of lessons and activities using the IWB that were used in this research. The second section includes a description of the research design, context, and participants. Also included are the IWB based activities, data collection and analysis. A description and discussion of the study’s trustworthiness is provided.

Researcher Background

I have worked in education for nine years as a teacher and administrator in the United States and the United Arab Emirates. During the time of this study, I taught grade three and had worked in the same district for five years. As a third grade teacher, I was tasked with implementing creative and meaningful lessons that integrated technology and met student learning outcomes. I was responsible for planning the English lessons for my grade level. I have a Bachelor of Arts degree in History and a Master of Education degree in Middle Grades Education. I have a strong belief that technology can help second language learners to acquire language. In my own class, I use IWBs during my instruction and have found them to be effective in assisting students with motivation and engagement. I have also personally benefited from learning with technology online through taking courses in my doctoral program. I feel that the integration of technology can help all students from elementary to university levels.

In my previous role as a Head of Faculty in a school in Abu Dhabi, I consistently observed many of my teachers struggling with student motivation in their classrooms when teaching English. Many teachers came to me for assistance, and for this I focused on using technology enhanced language learning (TELL) software in an action research project in order to study engagement and motivation in my qualifying examination project (Martin, 2012). A technology readily available in Abu Dhabi classrooms that can be used in a similar manner to the language learning software to motivate students was the IWB. In my role as a third-grade teacher, I found that many of my students are lacking in their English vocabulary, particularly their basic interpersonal communicative skills (BICS), which affected their acquisition of content such as mathematics and

science (Cummins, 1981). For these reasons, I was interested in studying ways how an IWB can be used in an ELL classroom for vocabulary development.

While conducting this study I taught grade 3. I was responsible for planning the literacy lessons for my grade level, therefore I began by designing the lessons to be used in this study. In addition to this I was responsible for implementing a variation of these lessons in my own classroom. I designed the lessons around research based theories and principles in second language acquisition, instructional design, and IWB features. about using IWBs and technology in classrooms, designing the lessons based on research and theory was extremely important. The lesson design utilized research based on comprehensible input and explicit vocabulary instruction for second language acquisition. The educational technology aspect included research based on IWBs and multimodal learning. After designing the lessons I sought input from the teachers that participated in this study to reduce subjectivity. I also kept a research journal throughout the process – from lesson design to the coding process and the writing of this dissertation - which helped me stay aware of any biases that might have influenced my research.

The lesson outline included learning standards, key vocabulary, materials, main lesson, guided/small group instruction, individual practice, and closing (Short, Echevarria, & Richards-Tutor, 2011). IWB instructional design principles and second language acquisition theory influenced the design of the lessons. The following section outlines the lesson design that informed this study.

Lesson Design

The lesson design for this dissertation utilized features that supported instructional content through research-based practices that were beneficial for English

language learners (Short, Echevarria, & Richards-Tutor, 2011). Vocabulary in this dissertation focused on Basic Interpersonal Communicative Skills (BICS) (Beck et al., 2002). The lessons are described here as they were first designed. They were later changed according to information received from the pre- observation interview and the post-first observation interview with the teachers.

Whole-class teaching was used with the IWB to demonstrate software, present and discuss information, and, with small group instruction, to enable students to show and receive feedback on their work (Levy, 2002). The use of multiple modalities such as audio and video was also incorporated in the lesson (Coyle, Yanez, & Verdu, 2010; Echevarria, Short, & Vogt, 2008; Hur & Suh, 2012, Vidot, 2011, Wood & Ashfield, 2008). Visual scaffolding where words are seen and heard in both English and Arabic were included in the lesson (Short, Echevarria, & Richards-Tutor, 2011). Students were also given the opportunity to manipulate items on the board (Interactive Technologies in Language Teaching, 2011).

Software for the Activities

The software that was used in combination with the IWB were Easy Interactive Tools©, PM Writing© IWB software, Microsoft PowerPoint©, Memory Game Maker©, and Sentence Game Maker©. Each of these programs were used in my school and utilized different features of the IWB that will be helpful when researching vocabulary development and were described in detail in this section.

Easy Interactive Tools©

Easy Interactive Tools© was the software that comes pre- installed with the IWB (Seiko Epson Corporation, 2011). The functions that were used were the annotation mode, which allowed the teacher/students to make changes to the displayed screen

using a highlighter, pen, shape, and eraser buttons (Figure D-1 and Figure D-2). These features allowed the teacher/student to make annotations on the screen with the IWB marker (Seiko Epson Corporation, 2011). The whiteboard mode allowed the teacher to upload an image to make changes. An example was included in Figure D-3, and this tool was useful when implementing a matching activity.

PM Writing© IWB Software

PM Writing© IWB software included an interactive big book with pictures, vocabulary, writing exemplars and text highlighting (Nelley et al., 2007). The information was presented in an easy-to-use manner for teachers and students. This program was used in combination with Easy Interactive Tools©. The PM Writing© IWB software features visuals for learning vocabulary and can be used with the Easy Interactive Tools© highlighter, pen, and shape buttons to teach new vocabulary to students. An example screenshot is displayed in Figure D-4. The PM Writing© IWB software also allowed the teacher to highlight vocabulary such as nouns, adjectives, and verbs, which is shown in Figure D-5.

Microsoft PowerPoint©

Microsoft PowerPoint© was used to introduce students to new vocabulary, as well as to match the sounds in English and Arabic. The primary feature was to introduce focus on specific vocabulary and to display the images through a slideshow (Microsoft Corporation, 2014). This software included transitions, animations and sounds in the activity. A sample screenshot is displayed in Figure D-6.

Memory Game Maker© and Sentence Game Maker©

Memory Game Maker© and Sentence Game Maker© allowed teachers to create multimedia language games to practice using language with interactive tasks and was

used with the IWB. These pieces of software were designed by the Languages Online Team, which is a part of the State Government of Victoria in Australia (Languages Online Team, 2008).

Memory Game Maker© is an interactive memory game that requires the students to find matching pairs of cards (Languages Online Team, 2008). This allows the teacher to insert text, pictures, and sounds in multiple languages. Example screenshots are included in Figure D-7 and Figure D-8. Sentence Game Maker© (Figure D-9) required the students to complete a sentence by selecting from a number of possible answers (Figure D-10). The game allowed the students to either drag and drop words or shoot words by interacting with the IWB and marker. This game focused on sentence structure and the order of words (Languages Online Team, 2008).

Second language acquisition theory impacted the design of the lesson through learning the meaning of new words (Beck et al., 2002; Stahl & Nagy, 2006), the presentation of word families (Edelen-Smith, 1997; Scheiner, 2005), associating words with images (Chun & Plass, 1999; Interactive Technologies in Language Teaching, 2011) and modification of input (Krashen, 1981).

The design of the lesson also incorporated appropriate speech, use of visuals and opportunities for discussion and concept clarification (Echevarria, Short, & Vogt, 2008). The IWB allowed for the use of visual input (Echevarria, Short, & Vogt, 2008), tactile input (Hall & Higgins, 2005; Schmid, 2008) and auditory learning (Plass et al., 2003). Table D-1 gives an overview of the software and activities that will be used in this study, along with the matching second-language acquisition and educational technology theories. The instructional design was based on the SIOP model, which incorporates

the following: learning standards, key vocabulary, materials, main lesson, guided/small group instruction, individual practice/assessment and closing.

Lessons and Activities

The first lesson went over a procedural text and had a focus on the nouns used within that text. The second lesson focused on the word families "–ing" and "–ug." Each teacher taught one lesson that was modified and changed after the first lesson and the post-first observation interview.

First Lesson-Making a Sock Puppet (Procedural Text)

During this lesson, students learned about how to make a Sock Puppet through reading a procedural text. The lesson was approximately 45 minutes and utilized whole-group, small group, and individual instruction. This lesson utilized the tactile, visual, and auditory aspects of the IWB (Bell, 2002; Chapman, 2005; Glover & Miller, 2002). The lesson also utilized speech that was at the majority of the student's proficiency level (Echevarria, Short, & Vogt, 2008).

Learning Standards

The learning standards included the ADEC Outcomes and the Language Learning Outcomes (Abu Dhabi Education Council, 2013; Echevarria & Graves, 2007). The ADEC outcomes were:

- Read words that include consonant letters that represent several different sounds or are silent.
- Recognize and use letters that represent the wide variety of vowels.
- Read 300 High Frequency Words
- Listen actively and engage in literary texts
- Recognize and use content words

The language outcomes were:

- The students will be able to read and identify the nouns presented in the “Making a Sock Puppet” text.
- The students will be able to follow the directions provided from the “Making a Sock Puppet” text to complete a project.

Key Vocabulary

The key vocabulary that was taught came directly from the nouns that were to be focused on in the exemplar text. The vocabulary was: “cat,” “puppet,” “sock,” “paper,” “pencil,” “scissors,” “glue,” “words,” “ears,” “whiskers,” “eyes,” “noise,” “hand,” and “fingers.”

Materials

This lesson used the Easy Interactive Tools©, PM Writing Software©, Microsoft PowerPoint©, and Memory Game Maker©. The lesson also used the materials to make a sock puppet, namely, a gray sock, black paper, yellow paper, pink paper, a pencil, scissors, and glue.

Main Lesson

The teacher introduced the vocabulary using Microsoft PowerPoint© (Figure D-11), and utilized an explicit vocabulary instruction method (Graves, 2006; Longo & Curtis). Each of the slides included vocabulary that students were required to know and that was spoken in English and Arabic (Echevarria, Short, & Vogt, 2008; Chapman, 2005).

After pre- teaching the vocabulary for the lesson as a whole class, the teacher read the story “A Sock Puppet” out loud (Nelley et al., 2007). After reading the text, the teacher encouraged several students to come to the board to see if they can recognize some of the vocabulary that was previously discussed during the PowerPoint© slides.

As the students came to the board, they highlighted the words using the feature from the Easy Interactive Tools© software (Ansell & Park, 2003; Bell, 2002). This gave the students an opportunity to understand the concept with the help of the teacher (Echevarria, Short, & Vogt, 2008). The main lesson was taught in order to help students to learn the meaning of new words (Beck et al., 2002; Stahl & Nagy, 2006).

Guided/Small Group Instruction

After completing the read aloud, the students completed their individual practice while several small groups were pulled to the IWB to complete an activity using the Memory Game Maker© (Wall et al., 2005). This activity allowed the students to use the IWB marker to match the vocabulary to the word (Ansell & Park, 2003). This activity utilized tactile and visual input to reinforce the vocabulary from the lesson (Interactive Technologies in Language Teaching, 2011). The small group instruction was used in order to further reinforce the meaning of the words.

Individual Practice/Assessment

The students completed the instructions to make a sock puppet from the story. When they were finished with the puppet, they practiced their oral language skills by reading the text to each other, as well as working on the practice sheet (Figure D-12).

Closing

At the close of the lesson, the teacher reviewed the vocabulary with the class by using the Sentence Matching Game Maker© (Figure D-13). The students were able to come to the IWB one at a time to see if they can fill in the blanks using the IWB marker and visual cues (Bell, 2002; Gray et al., 2007; Vidot, 2011).

Second Lesson-Poetry with Word Families (“-ing” and “–ug”)

This lesson focused on teaching the word families “–ing” and “–ug” through poetry. The students learned about rhyming words and how to switch beginning consonant sounds with others (Scheiner, 2005). The lesson had a focus on the tactile input feature of the IWB (Ansell & Park, 2003; Bell, 2002; Cuban, 2001). The students had an opportunity to clarify the concept of word families throughout the lesson (Echevarria & Graves, 2007).

Learning Standards

The learning standards included the ADEC Outcomes and the Language Learning Outcomes (Abu Dhabi Education Council, 2013; Echevarria & Graves, 2007). The ADEC outcomes were:

- Demonstrate understandings of word relationships in word meanings
- Recognize and use letters that represent the wide variety of vowels
- Read 300 High Frequency Words
- Use a large number of phonograms when spelling words
- Use frequently appearing short vowel patterns that appear in multi-syllable words

The planned language outcome was:

The students will be able to read and identify the “–ing” and the “–ug” word family

Key Vocabulary

The key vocabulary came from the “-ing” and “–ug” word families.

- sing, spring, wing, bring, zing, everything, swing
- bug, snug, rug, mug, hug

Materials

The lesson used Easy Interactive Tools©, Microsoft PowerPoint©, and word family poems (Microsoft Corporation, 2014; Seiko Epson Corporation, 2011; Scheiner, 2005). The two poems were titled “Sing Bird!” and a “Snug Bug!” (Scheiner, 2005).

Main Lesson

The teacher read the first two lines from the poem “Sing Bird!” and highlighted the words “sing” and “spring” to introduce the students to the word family “–ing.” This utilized the highlighting text feature from Easy Interactive Tools©. The teacher then invited one of the students to come to the board to see if they could spot and highlight the word “wing.” The teacher then invited students up one at a time to finish writing the five words: “bring,” “zing,” “everything,” and “swing.” This portion of the lesson focused on using the IWB with tactile input through highlighting and writing words (Ansell & Park, 2003; Bell, 2002).

Guided/Small Group Instruction

While the students completed their individual practice sheets, the teacher took the students to the IWB in small groups. The teacher then displayed the list of words and had the students put triangles around the “–ing” words and circles around the “–ug” words using the shapes feature from Easy Interactive Tools© (Figure D-14). This section of the lesson focused on interacting with the words through the presentation/circling of word families and through the reinforcement of vocabulary (Hall & Higgins, 2005).

Individual Practice/Assessment

The teacher wrote the word family “–ug” on the board, and elicited some response to see if they can come up with some words and give the students the poem “A Snug Bug!” to complete (Figure D-15) (Scheiner, 2005).

Closing

The teacher displayed the poem “A Snug Bug” and had the students come up to complete the words and highlight the word families (Figure D-16). Sample answers to the poem were “bug,” “snug,” “rug,” “mug,” and “hug.” The closing focused on ensuring that the students understood the concept of word families (Scheinner, 2005).

Methodology

Qualitative research methods were used in this exploratory study because they allowed me to “understand and interpret how the various participants in a social setting construct the world around them” (Glesne, 2006, p. 3). Qualitative research methods can be advantageous because of access to multiple perspectives of the participants, potential for spontaneity and open-ended interactions (Dooley, 2001). Contextualizing, understanding, and interpreting events that take place is possible through qualitative research. In this study, the lessons were first designed and following interviews with the teachers, changes to the lessons were made before a second implementation. Further, as following the interview and observations with one teacher, the lessons with the second teacher were also reviewed and minor changes made. The research was conducted with IRB approval from the University of Florida and both teachers signed consent forms (Appendix C) before participating in this study.

Context

This dissertation took place in an all-boys school in four third grade classrooms over 17 weeks in Abu Dhabi. The school had an enrollment of 980 students in first to fifth grades. The third grade classes had 24, 25, 23 and 25 students each. Students attended school from 7:15 a.m. to 1:00 p.m., five days per week. The literacy/English curriculum was designed by the Abu Dhabi Education Council and was intended to improve student learning outcomes through balanced and focused teaching. Each of the classrooms were equipped with two computers and an IWB. Eighty five percent of the students were local Emiratis, and the remaining were from Jordan, Egypt, Syria, or Western/Eastern European countries. The students took 35 instructional periods each week. Each instructional period was 45 minutes in length. Native English speakers taught 15 periods per week of English, mathematics and science. The remainder of the periods were taken in Arabic/Islamic studies, social studies, art, physical education, music, art and computer studies. A school day consisted of seven periods with a 20-minute break for lunch and outdoor play. IWBs were used in the classroom daily during whole-class and small-group instruction during English, math, and science.

Participants

The classes observed in this study were taught by licensed, certified teachers from the United Kingdom and Canada. The first teacher (Amy) is a British female who had five years of experience in elementary education in the United Kingdom, Uganda and the United Arab Emirates. She had some experience using IWBs in her instruction. She had a Bachelor of Arts degree in Education from a university in the United Kingdom. The second teacher (Bob) was a Canadian male who had 10 years of experience in elementary education in Canada and the United Arab Emirates. He also

had a Bachelor of Arts degree in Education and a Master of Education degree in Educational Technology from a Canadian university. He had significant experience using an IWB in his instruction. These teachers were asked to sign an informed consent form, which is included in Appendix C.

The participants were students in four classes (Class A-1, Class A-1, Class B-1, and Class B-2) of Grade 3 ELL students (N = 97). All of the students were male and between the ages of 6 and 8. Many of the students' English skills are not reinforced at home because the native language is Arabic, and for many there is limited English language use in the household. The majority of the English instruction that the students received was in the classrooms with native English speakers. The students have used the IWB before in previous lessons for the previous three years. Participants' identities were kept confidential when collecting, analyzing, and writing up the findings.

General Procedures

I gave a 30-minute presentation on the details of this study to the two teachers that are participating. The lesson activity design utilized theories based on second-language acquisition, comprehensible input, vocabulary instruction, and IWB techniques (Beck et al., 2002; Echevarria, Short, & Vogt, 2008; Krashen & Terrell, 1983; Plass et al., 2003). These theories and designs were discussed with the teachers during the presentation. During this presentation, I reviewed the theoretical framework, interview process and observation details. We also discussed the lesson that we used, and I took questions. The teachers received a summary of the presentation and study, printouts and electronic copies of the lesson and materials, reflection documents and an informed consent form. On my team, I was responsible for planning and collecting resources for

literacy; therefore, I provided the materials that the curriculum requires that we use. The teachers had choices on how they choose to use the materials in their classrooms.

After the presentation, the teachers had three days to reflect on the lesson and come back with any changes that they believed would improve the lesson and vocabulary development using the IWB. We discussed these changes about using the IWB during the pre-observation interview, which lasted approximately 60 minutes. We decided what changes or additions to make to the lessons at that point in time. The pre-observation interview was recorded with the permission of the teachers.

I visited their classrooms to observe them within one week of conducting the interview. After completing the lesson and observation, I conducted a post-first observation interview with them within two days, which was also recorded with their permission. Changes recommended by the teachers from these observations and interviews were made to the lessons and materials for further implementation in the follow up lessons. I observed the follow up lesson and conducted a post-second observation interview, which followed the same procedure. Both post observations focused on reflecting on the IWB activities and usage in the classroom, activity and lesson design, feedback from the teacher and recommended changes. The interview protocols are included in Appendix A. A study implementation schedule is included in Table D-2.

The data from the pre-observation interviews, observations, post-first observation interviews and post-second observation teacher interviews were analyzed using qualitative techniques (Dana & Yendol-Hoppey, 2009; Dooley, 2001; Glesne, 2006). Figure D-17 provides a graphic overview of the data collection process.

Research Design

Exploratory research that is qualitative in nature refers to “broad-ranging, intentional, systematic data collection designed to maximize discovery of generalizations based on description and direct understanding of an area of social or psychological life” (Stubbens, 2008, p. 328). This study attempted to understand how IWBs could be utilized in classrooms for ELL vocabulary development. According to Stebbins (2008), the primary goal of exploratory research is “the production of inductively derived generalizations about the group, process, activity, or situation under study” (p. 329).

An exploratory, qualitative research design was used for this study (Figure D-18). In this research design, qualitative data was collected first through teacher pre-interviews and classroom observations (Glesne, 2006). Additional data collection and analysis of the qualitative data continued during the second part of the study through follow-up classroom observations, post-lesson teacher interviews and a reflective journal. This provided a detailed description of the use of IWBs in the classroom, as well as thoughts and ideas about vocabulary development. The qualitative data was collected through classroom observations and teacher interviews. The results were then analyzed and interpreted to understand how IWBs can assist with vocabulary development (Glesne, 2006; Stubbens, 2008).

Data Collection

Data was collected through individual teacher interviews, classroom observations, recording of the lesson and a reflective journal. The combination of these data sources helped to answer the research question of how an IWB can be used to

facilitate vocabulary development. The data was transcribed, compiled, coded and analyzed to identify how an IWB can be used to facilitate vocabulary development.

Teacher Interviews

For the purposes of this research project, I interviewed both teachers three times "in search of opinions, perceptions, and attitudes" (Glesne, 2006, p. 80) toward use of the IWB and vocabulary development of their students. Pseudonyms were used in writing the results in order to protect the participant's confidentiality. Three interviews were conducted before the first lesson, after the first lesson and after the second lesson. The style of interviewing that was used was semi-structured, which consisted "of a set of questions carefully worded and arranged for the purpose of taking each respondent through the same sequence and asking each respondent the same questions with essentially the same words" (Patton, 1987, p. 112). Although flexibility in this type of question is somewhat limited, potential for bias was reduced due to the systematic nature of the interview. These interviews had a mix of structured, open-ended and specifically placed probing questions. The teacher interviews were intended to provide a rich collection of data in thinking about ways to implement IWB-based instruction in an ELL classroom and was audio-recorded and subsequently transcribed. Each of the interviews with Amy lasted 15, 22, and 21 minutes, respectively. Each of the interviews with Bob lasted 20, 24, and 28 minutes, respectively. I interviewed Amy in her classroom and in the media center at our school. I interviewed Bob in his classroom and my classroom.

The pre-observation interview focused on teacher background knowledge about using an IWB, previous knowledge about second language acquisition theory, vocabulary instruction methods in their classroom and feedback on the IWB-based

lesson that was provided during the presentation on the study. The post-first observation interview followed the first lesson observation and focused on the lesson feedback. The post-second observation interview focused on lesson feedback and teacher reflections about the implementation of IWB based lessons. The three interview protocols are included in Appendix A.

Classroom Observations

The four lessons were observed and documented using an adaptation of portions of the SIOP Model and the International Society for Technology in Education's (ISTE) classroom observation tool (ICOT) (Echevarria et al., 2000; Fledderus et al., 2011; ISTE, 2013). This tool that I created focused on second language acquisition and educational technology theories through its focus on production, self-correction, the opportunity to engage in high levels of target language input, modeling, the use of visuals, opportunities for discussion and concept clarification, role of the teacher and timed usage of technology. The tool also had a section for writing about the conclusion, as well as suggestions. This is included in Appendix B.

The portion that was adapted from the ICOT by ISTE has a focus on timed usage of technological tools such as an IWB in application as well as notes on student interaction (ISTE, 2013). This information was studied and observed during the mini-lesson, small-group instruction and closing. The portions that were adapted from the SIOP model focus on the level of usage of the supplementary activities with the IWB and how the materials make language input comprehensible (Echevarria, Vogt, & Short, 2000; Krashen, 1983).

Recording of IWB Interactions

The student and teacher interaction with the board was recorded using visual capture software. This software was turned on during the entire lesson and was saved on the computer. The recording focused on the mini lesson, small group and closing aspects of the lesson. This recoding was analyzed for ways that language is made comprehensible and/or facilitates second-language development (English). The analyzed segments were selected for the interviews. The selections were used as a stimulus or recall for teachers to elaborate on their instructional decisions and views about using the IWB as a tool for language development.

Reflective Journal

This journal was used in order to record my thoughts during a) the planning phase, b) before and after the interviews and classroom observations and c) during data analysis. According to Glesne (2006), “this is the time to write down feelings, work out problems, jot down ideas and impressions, clarify earlier interpretations, speculate about what is going on and make flexible short- and long term plans” (p. 59). The journal contained three sections, which were curriculum development, reflection on classroom observations and how I decided to research the lesson so as to account for biases. It was important that I document any bias or attitude that influences data collection and analysis. The journal also included space for reflections about emerging patterns and ideas within the data (Glesne, 2006; Saldana, 2013). The notes were transferred to a word processing journal for later review and analysis.

Data Analysis

Exploratory data analysis gave me as much room as possible to find new concepts. After I finished the data collection process, I “closely examined and further

compared with an eye, first, to discovering their common properties and, second, to melding these properties in concepts and generalizations about the area of social life under study" (Stebbins, 2008, p. 330). This organization allowed me to find and discover new relationships. The data that was collected and transcribed, and then the data chunks were given themes and descriptions.

This dissertation used thematic analysis for analyzing the qualitative data. Braun and Clarke (2006) defined thematic analysis as a "method for identifying, analyzing and reporting patterns (themes) within data," and it "minimally organizes and describes your data set in (rich) detail" (p. 6). Thematic analysis provided clarity about meanings of the raw data. This type of analysis "involves searching across a data set" in order "to find repeated patterns of meaning" (p. 15).

Analysis was conducted using data from the teacher interviews, classroom observations, including the observation tools as well as the samples selected for teacher interviews (as stimuli), recording of IWB interactions and my reflective journal. This process required "meticulous attention to language and deep reflection on the emergent patterns and meanings" within data (Saldana, 2013, p. 10). The screenshots from the recording of the lessons supplemented the classroom observations, and my reflective journal and was used during teacher interviews. The reflective journal assisted in the whole process, and my notes were analyzed to supplement the existing data and identify my bias as a researcher embedded in the context.

Coding Process

This study attempted to understand how IWBs could be utilized in classrooms for ELL vocabulary development. The data was analyzed and interpreted through thematic analysis (Braun & Clarke, 2006). Data from interviews, observations, and personal

reflections were thoroughly read and analyzed several times then deconstructed. I utilized the highlighting and comment tools in Microsoft Word in order to code and re-code the data on the interviews, observations, and research journal. The first analysis produced 223 initial phrases that were then analyzed to produce primary codes and sub-codes of the data after a second thorough analysis. These primary and sub-codes were then applied to the observations, interviews, and reflection notes in a second level of analysis displayed in Tables D-3 through D-5. From further analysis three themes emerged around vocabulary development and the IWB, which included multiple uses of the IWB for ELL students. Figure D-19 provides a visual representation of how the themes and corresponding codes overlap.

Trustworthiness of the Study

The objective of this study was to research how IWB-based lessons and activities were implemented in ELL classrooms for vocabulary development. This study used qualitative data in order to understand and answer the research question through the use of teacher interviews, classroom observations and a reflective journal. According to Glesne (2006), persistent observation, triangulation of data sources, clarification of researcher bias and rich descriptions of the subjects all go help with trustworthiness. This study addressed transferability, dependability, credibility and limitations in order to further validate the study's trustworthiness.

Transferability

Making connections from this study to other similar schools and contexts raises questions about transferability. The findings of this study can be applied in other classrooms in Abu Dhabi with similar student populations and class sizes. Although it may be difficult to apply this study to other countries, ELL students and class sizes, a

rich description of the place where the study takes place along with the participants allows other teachers and researchers to determine if the study is transferable to their particular context (Dooley, 2001).

Dependability

One of the issues in qualitative studies is the number of variables within a particular environment (Dooley, 2001). I accounted for these variables and the dependability of this study by providing a rich description of the methodology used in the study through the use of data collection and data analysis.

Credibility

Credibility refers to the validity of the outcomes of the study. If the codes and themes that emerge from the data are believable, then the study has a high level of credibility. In order to account for this, I established a level of rapport with the teachers on my team through interviews and cooperative planning. I also have established relationships with the students through cooperative teaching and engagement. My team uses a similar teaching and planning structure; therefore, I was familiar with the outline of the instruction in the classroom. I also observed the same lessons in four different classrooms from the two teachers, which will ensure that all outcomes are reached (Glesne, 2006). I shared my classroom observations with the teachers, increasing credibility of the data because the teachers corroborated or negated what was observed. I shared my final results with both teachers. I also provided descriptions of the codes.

Limitations

The limitations of the study included focusing solely on the usage of the IWB for students' vocabulary development where there could be other variables that facilitate

student vocabulary development in the classroom. Additionally, the data that is collected from multiple data sources could pose a potential bias as the data is subjective. One other limitation of the study was the sample size of the four classrooms, which may or may not reflect the ability of all students in the school. Researcher bias must also be recognized. I have had positive experiences during my qualifying research project and believe that well-designed IWB materials can assist with vocabulary development. Some of these factors may affect my research and data analysis. By keeping a reflective journal, as well as having an outside researcher help with coding the data, I attempted to limit some of the effects of bias.

CHAPTER 4
RESULTS

This chapter reports the results of the data obtained through thematic analysis in qualitative research (Braun & Clarke, 2006) during observations of the use of Interactive Whiteboards (IWB) in four ELL classes in the United Arab Emirates. This chapter describes how the IWB was used to facilitate vocabulary development in four ELL classrooms in the United Arab Emirates. This study investigates the following question: How can IWBs be used to facilitate vocabulary development in an ELL classroom? The study took place in four third-grade classrooms taught by two native English-speaking teachers. Data from the study were obtained from teaching interviews, classroom observations, a reflective journal, and screenshot recordings of the lessons. Details of classroom observations and changes made to the lesson design are provided in Appendix E.

Background information about the teachers is first provided, which details their experiences with using the IWB and teaching ELL students. The three major themes that emerged from the data are then presented in this chapter. They were:

- Tactile uses of the IWBs: “Where they can come up and use the pen and markers.”
- IWBs and audio learning: “Vocabulary building is hearing the words over and over again.”
- IWBs and visual learning: “You might associate a word with a picture.”

Teacher Profiles

I conducted a pre-observation interview with both teachers about their backgrounds, knowledge and experiences with respect to using an IWB and vocabulary instruction methods in their classrooms, also getting their feedback on the IWB-based

lesson. This section describes the two teachers involved in this study and their prior experiences teaching vocabulary development with technology.

Amy

Amy taught in the United Kingdom for several years before coming to work in the United Arab Emirates. In the United Kingdom, she predominantly taught students from Pakistan and India, so she had experience working with students that were learning English as a second language. She taught English as a second language in Uganda for a year. She has taught in the United Arab Emirates for two years and has experience working with Arabic-speaking students from the Middle East. She spoke about the abundance of resources available in the United Kingdom for English as second-language learners. Her experience teaching English as an Additional Language (EAL) in her home country was similar to teaching ELL in the United Arab Emirates.

With respect to the use of technology in the classroom, Amy was very insistent about receiving adequate training and felt that using technology would create a better lesson in the classroom for her students. In her class, she recorded many different things with the students. She used a radio, CD player and computer. She also incorporated different types of interactive websites based around science and math in her practice. She stated she used Visual Mathematics, Smart Notebook, and Google Documents. Amy used video to teach action and songs, as well as spoke about using a radio or CD player. She also implemented camera-based activities in her classroom.

According to Amy, in contrast to the United Kingdom, the United Arab Emirates does not have many resources available for teaching English as a second language; therefore, it is necessary to be creative. She described her experience teaching in the United Arab Emirates as challenging because she said she was not specifically trained

to work with English as a second-language student. When teaching vocabulary, she spoke about the difficulty of finding resources for the IWB when using PowerPoint. She said she has used many similar techniques in her home country as in the United Arab Emirates. She spoke about using SDAIE (Specially Designed Academic Instruction in English) strategies, which are ESL-specific strategies as well as implementing a balanced literacy program. She also spoke about utilizing roles, realia, TPR (Total Physical Recall) strategies which focus on acting out vocabulary.

In our school, the literacy program that we use is balanced literacy. Amy elaborated on the different strategies that she used in her classroom. She also spoke about pre-teaching the vocabulary ahead of time so that the students were aware of the words before the lessons. She was very concerned about teaching the words to the students in context so that they would be able to use the words. She also had experience using PowerPoint to display words and pictures to the students. She liked that the students used the vocabulary in different contexts. She felt that using technology provided her with better lessons for her students. She also felt that boys liked to work with hands-on activities, and she looked forward to using additional tools to support her teaching-student learning.

Amy felt that the IWB assisted with modeling handwriting, the formation of letters and reading Arabic words. Word association with pictures and vocabulary, as well as scaffolding language through modeling handwriting, were also two areas Amy felt the IWB was helpful in the planning stage. She associated words with pictures and felt it was helpful with teaching science and different academic subjects, as well as teaching with PowerPoint.

Bob

Bob is originally from Canada, where he taught ESL in Vancouver and Toronto. He started teaching English as a second language in Japan 15 years ago and has taught for five years in the United Arab Emirates. He has taught adolescent and adult students. He has a master's degree in educational technology from a Canadian University in which he studied technology applications and ESL students, although he reflected that in between that time, many things have changed as far as technology in the classroom. Changes in technology have transformed how he teaches ESL students.

Bob has taught vocabulary in his classroom in several different ways. He has pre-taught the vocabulary before reading a book, used a dual language dictionary in English and Arabic, used PowerPoint because of its perceived benefit of teaching vocabulary through using visuals in the teaching environment, used a program called Visual Words with PowerPoint and also used the technique of associating a word with a picture that he felt was helpful when teaching an English, math or science lesson.

I asked him about his prior usage of IWBs in his classroom and teaching practice. He found the visual aspect of IWBs assisted with teaching and learning but also assisted students that did not necessarily learn through visual techniques. He stated:

> All learners don't necessarily learn visually. The IWB has helped my teaching. When something is projected on the whiteboard, it gives access to that on the screen. It actually really helps the students understand because they can relate to it much better than on paper and pencil. (Bob, interview, May 31, 2014)

Bob found it useful that students can use the pen and markers because the "kids really enjoy that, and they feel successful" (Bob, interview, April 29, 2014). He also made the argument that the students felt successful using IWBs because they could

make mistakes and not feel that they did not succeed. He said, “They can access the technology and show their other classmates what they are doing and have success as a class. They enjoy taking turns, and they enjoy using the technology” (Bob, interview, April 29, 2014). At the same time, he had experienced technical challenges in having the students use the IWB markers. He stated the markers did not work at times, and students had to push down harder. He made the argument that having students work together would help overcome some of the challenges.

One of the programs that Bob used with IWBs was SmartNotebook. He used this program to have the students manipulate objects, reveal sentences and give them access to many different pictures. He stated, “With this particular program, you have the kids come up and erase things” (Bob, interview, April 29, 2014). He also used the IWB to model handwriting and to scaffold learning. The following sections cover the major themes that emerged from thematic analysis.

Tactile Uses of the IWB: “Where They Can Come Up and Use the Pen and Markers”

Tactile uses of the IWB consisted of occasions when teachers and students used various IWB markers, namely to manipulate vocabulary, for teacher modeling, in implementing learning strategies and for assessment strategies. Using the IWB markers for manipulating vocabulary helped with assisting students with their vocabulary development. Both teachers were able to make assessments in their notebooks from seeing the students complete their assignments on the IWB. Data from the study provided evidence that using the IWB marker for teacher modeling was also helpful with vocabulary development. Several learning strategies, such as previewing, highlighting vocabulary, using the words in context, utilizing the drag and drop feature, making

mistakes and reviewing vocabulary, were effective methods for developing students' vocabularies through the usage of the IWB markers. These strategies were effective for second-language acquisition because students needed opportunities to practice using language through input-rich activities. The IWB markers were also effective in assisting the teacher with implementing assessment strategies. Figures 4-1 and 4-2 display screenshots of two of the tactile features of the IWB. Figure 4-1 displays a student using the whiteboard marker to match vocabulary with the picture. Figure 4-2 demonstrates how the teacher used the IWB markers for highlight vocabulary and model.

Manipulating Vocabulary/Using IWB Markers

The IWB markers assisted with making the lessons more tactile and hands-on for the students. Many ELL learners benefited from the use of tactile learning strategies because they did not have the language to comprehend English vocabulary at a high level, so they must had been given opportunities to manipulate and touch. Comprehensible input requires that students have opportunities to use language at a level that scaffolded (or assisted) but that is slightly more difficult than what they can understand on their own. In this study, using the IWB markers provided students with the opportunity to practice manipulating the English language at a higher level. Both teachers in this study felt the tactile aspect of the IWB helped with vocabulary development, particularly in this environment where there all boys who they felt were more generally kinesthetic and tactile learners. Amy stated, "I work in a boys' school, and most of my boys like things that are hands-on" (Amy, interview, March 13, 2014). Bob was more specific:

> You could see where there was an activity where they [the students] could write on the board. Where they can manipulate things on the board. They were writing it or circling it. They enjoyed that. I think that's really what the

> IWB offers for a class. If you have engaging activities, it really does appeal to the tactile learner, which is one of the benefits of the Interactive Whiteboard. (Bob, interview, May 7, 2014)

Both teachers felt that the tactile aspect of the IWB was useful for helping male students with vocabulary development. Amy stated, “I think that when you are working with boys. Boys are a bit more hands-on, and they like things that are ‘gadgety.’ They require things that don’t just have them sitting at a desk and just writing. I do not find that my boys enjoy that” (Amy, interview, April 1, 2014). In both teachers’ opinions, they felt using the IWB would get the best out of their students through the interaction with vocabulary. Explicit vocabulary instruction gives students the opportunity to think about and use words through exposure to targeted vocabulary.

Bob made comments about students using the markers when he stated, “It also allows the students to manipulate items through the use of the IWB markers” (Bob, interview, May 31, 2014). By using the IWB markers, students were able to use “IWB markers to manipulate different words and pictures for vocabulary development” (Bob, interview, May 31, 2014). The word manipulate came up consistently when discussing development of vocabulary. For instance, Bob noted, “As far as developing their vocabulary, the IWB allows them [the students] to see, hear and manipulate the words, which goes a long way in assisting them with usage of the vocabulary (Bob, interview, May 31, 2014). Through manipulating vocabulary, students were able to modify their input, which allowed them to make changes immediately, which helped ELL students with vocabulary development. This was one of the central components of comprehensible input.

The primary method of interacting with the IWB was through the markers; therefore, the teachers felt students gained much more in terms of vocabulary

development through completing the activities on the board. For instance, in Amy's second interview, she stated, "I think that the more interaction then the more the children are able to sit back and observe what was going on. They can learn from their friends as well. I thought that it supplemented it. It made the learning more meaningful (Amy, interview, March 22, 2014). The IWB allowed students to manipulate and interact with the lessons and curriculum in a way that they could not with pencil and paper. Students who were at the beginning level of language and vocabulary acquisition might not have been able to orally use the language; therefore, using the IWB markers allowed them to get practice with vocabulary without speaking. For instance, students were able to manipulate sentence and sentence structure. They were also able to correct their own mistakes, which they might not have been able to do if required to orally use the language. The other students in the classroom were able to view the structure of the sentences as well. The students in Amy's class were able to drag and drop sentences and put the words in correct syntactic order. They were given longer sentences, and when they made a mistake, they corrected their own mistake.

Manipulating the vocabulary allowed the students to think about what they were learning rather than what they were doing (Amy, interview, April 1, 2014). The IWB supported their vocabulary development by allowing the students to build vocabulary through making complete sentences rather than focusing on the physical aspect of writing (Figure 4-3). Bob felt that the IWB was "very effective developing the vocabulary of my students. The fact that it is very hands-on and intuitive allows the students to have opportunities to manipulate different objects, which allows them to develop their vocabulary" (Bob, interview, May 7, 2014). ELL students were given multiple

opportunities to practice with vocabulary, recall words and acquire language. Although this is not different than having the students cut and paste with paper, the teachers reported they felt the students greatly benefited from using the IWB markers to manipulate the sentences.

Teacher Modeling

The IWB markers were very useful for the teachers to model how to use the vocabulary in the classroom. During Amy's second lesson, she read through the story and gave directions one time, and then she had the students come up and highlight what they did. The goal was to make a cat puppet with a sock. The language outcome was to read and identify the nouns presented in the 'Making a Sock Puppet' text. After presenting this, the students came up and took turns highlighting the words. The words "cat," "puppet," "sock," "paper," "pencil," "scissors," "glue," "words," "ears," "whiskers," "eyes," "noise," "hand" and "fingers" were identified. According to Stahl and Fairbanks (1986), there are three tiers of words that are required to communicate in English. The words that were selected for this activity were primarily tier-one words, which were necessary for beginning-level ELLs. Tier-one words are words that are used frequently in the English language, whereas tier two and tier three are more academically focused. By having an understanding of the vocabulary needed for the activity, the students were able to complete the assignment (Figure 4-4).

By using the markers, the teachers modeled how to identify vocabulary. Bob was able to model how to highlight different words using the IWB markers. Bob modeled how to write the word family in the activity. Word families help students to understand and read patterns of letters in a fluent manner, which helps to build access to words that assist ELL students with vocabulary development. During the activity, Bob's students

were able to come up to the board and successfully circle the first word, “rug,” for the word family “–up,” which was evident during the lesson (Figure 4-5). Bob was also able to model how to write words on the board, as well as emphasize endings of the word families. The students then practiced this when they came up to the board, which facilitated vocabulary development.

Implementing Learning Strategies

The IWB markers were also used to highlight vocabulary. In Amy’s first lesson, she displayed the instructions for making a sock puppet and explicitly asked the students if they could highlight some of the words that were just discussed. She then had the students highlight the words, which helped them with their vocabulary development. When they highlighted the word “whiskers,” Amy prompted the students to orally repeat the word (Figure 4-6). As they highlighted the words, Amy also had them say the word in Arabic and English.

The highlighting technique was also effective when previewing the text by discussing vocabulary before reading the text through the PowerPoint. Bob also felt the highlighting and tactile part of his lesson was effective because the students could access and use the words (Figure 4-7). Explicit vocabulary instruction exposed students to words in context, and the ELL students benefitted from using the highlight tool to directly discuss the vocabulary.

In Bob’s lesson, he invited students to come up to the front to highlight the word family “–ing.” The language objective was to be able to read and identify the “-ing” and the “-ug” word families. Bob felt that when using the highlight tool, “As a whole class activity, it really does highlight what you are trying to teach in the objective for the lesson, and when the students actually saw and were able to highlight the words that

we were using, I think it was really effective (Bob, interview, May 7, 2014). Using the highlight tool helped students to identify vocabulary, which assisted with vocabulary development through giving students an opportunity to manipulate the words. Another learning strategy that was implemented through tactile methods was allowing the students to use the words in context. This helped the students to understand what the words actually meant, which enhanced the students' comprehension and vocabulary development. Amy stated the IWB allowed the students "to interact and to have an image readily available, and [it] allows them to practice orally and to manipulate the vocabulary" (Amy, interview, March 22, 2014). The students were also able to create simple and complex sentences, which required them to have an understanding of the vocabulary (Amy, interview, April 1, 2014).

A feature that was evident was the drag and drop tool. Through the use of the drag and drop tool, the students put the vocabulary in place through making a connection between the IWB activity and the sock puppet activity. Amy stated, "If we did not have the PowerPoint activity, then they would not have achieved the aim of the activity" (Amy, interview, March 22, 2014). Another tactile aspect of the IWB was that it allowed the teacher to review the vocabulary at the end of the lesson. During Bob's first and second lessons, he was able to place the story "A Snug Bug" on the board and then have the students come up to the board to write the endings of the words using the IWB marker.

Assessment Strategies

Having the students use the IWB markers allowed the teacher to conduct informal assessments of students. Amy stated, "Sometimes you just want to see a quick assessment of what they know. It was not about writing or reading but that [they]

understood the vocabulary" (Amy, interview, March 22, 2014). The highlighting feature allowed Amy to informally assess the students. She also stated, "Dragging to create the sentences I was able to see through my teacher assessment that these boys know the meaning of these words and they are fully aware of the activity and what is expected of them" (Amy, interview, April 1, 2014). She could see the students knew how to ask their friends if they did not know a particular word. They were able to use the marker to interact with the vocabulary in a tactile manner to assess if the student's vocabulary was extended and developed. Amy was able to assess students through visually monitoring them. She made notes in her notebook, which she stated would help her with remediation at a later time.

In Bob's lesson, he stated that he "could tell when students actually know what they were looking for, and it helped with my assessment" (Bob, interview, May 31, 2014). Further, he was "able to make quick assessments of the students, which will help me when I plan lessons" (Bob, interview, May 31, 2014). Using the IWB marker allowed the teachers to determine if the students understood the vocabulary, which could assist with planning future lessons for vocabulary development. The next theme focuses on IWBs and audio learning for vocabulary development.

IWBs and Audio Learning: "Vocabulary Building Is Hearing The Words Over And Over Again."

The IWB supported vocabulary developing through audio output and oral input. The audio aspect of the IWB was effective in supporting bi-literacy, implementing learning strategies and teacher modeling. Bi-literacy was supported, which allowed the students to use both their native Arabic and second-language English skills to develop their vocabularies. The IWB also allowed the teachers to model correct oral usage of

vocabulary. Learning strategies like differentiation and repetition were also effective in developing vocabulary. The oral aspect of the IWB was also useful in assessing students on their knowledge of vocabulary.

Supporting Bi-literacy

Having an understanding of the vocabulary in English and Arabic is one of the overall goals of the academic program in Abu Dhabi. Building vocabulary supports this goal in both of the lessons. Amy's lesson focused on using the words out loud in Arabic and English. In her lesson, she was able to encourage the students to listen and use the vocabulary in English and Arabic. She wanted to know if the PowerPoint, "read the Arabic words as well? I just wondered that if the children were to hear the word sock if there was time for them to hear and say the word" (Amy, interview, March 13, 2014). A delay was built into PowerPoint whereby students were able to hear the vocabulary in English and Arabic. Amy reminded the students to say the word in English and Arabic when they saw the picture (Figure 4-8).

They could have also asked questions about the meaning of the word and discussed among themselves in English and Arabic. Amy felt, "The IWB allows you not only to build their English vocabulary but also their vocabulary in Arabic to make meaning of the actual words" (Amy, interview, April 1, 2014). During Amy's second lesson, she decided to change her strategy and have the students read the word and say it in English and Arabic.

Teacher Modeling

Pronunciation of the words through teacher modeling helped students in their oral acquisitions of the vocabulary. Amy modeled how to say the words and demonstrated how to make a sock puppet. There are many different pronunciations of

words in English and Arabic. Having teacher interaction and teacher input assisted with student learning and vocabulary development. Bob stated, “Developing their vocabulary with the IWB allows them to see, hear and manipulate the words, which goes a long way in assisting them with usage of the vocabulary” (Bob, interview, May 31, 2014). Both teachers felt the audio and oral aspect of the IWB allowed them to model how to properly use the vocabulary, which provided high quality input for the students.

Implementing Learning Strategies

The audio aspect of the IWB lends itself to several learning strategies, such as differentiation, repetition and displaying/hearing vocabulary. Many students learn through oral methods rather than visual or tactile. For instance, Amy stated, “If I had a child that was in the classroom that was deaf, they [sic] could see the images and the words. If I had a child that was blind, they [sic] could actually hear the sound” (Amy, interview, March 22, 2014). This aspect allowed the teachers to differentiate the lesson through playing audio. Having this feature allowed the teachers to cater to the learning needs of all of the students. Students needed to hear the actual words to build their vocabulary.

Both teachers implemented and stated that repetition was important for vocabulary development. During Bob’s lesson, he repeated the words several times to students during his mini lesson. Amy stated that it was important to “create sounds with it because for me vocabulary building is hearing the words over and over again” (Amy, interview, April 1, 2014). Students needed to repeatedly hear and use the vocabulary in order for development to take place. By using the audio aspect, the teachers were able to encourage students to “say the word because sometimes in your head it sounds different than if it is coming off of your tongue” (Amy, interview, April 1, 2014). The audio

aspect of the IWB supported student vocabulary development through encouraging repetition and oral practice of the words. Although this is similar to student repetition in a non-IWB classroom, the teachers felt the combination of audio and visual aspects supported vocabulary development.

IWBs and Visual Learning: "You Might Associate A Word With A Picture."

The IWB supported visual learning through the projection display aspect. Several themes emerged, which included teacher modeling, implementing learning strategies and supporting bi-literacy. The teachers were able to model reading and using vocabulary through displaying pictures and text. Learning strategies, such as displaying visuals and pictures, using PowerPoint, word association, differentiation, displaying text and using the vocabulary in context, were evident in the lessons. The IWB visual aspect also supported bi-literacy. The display aspect of the IWB allowed pictures of the target words to be displayed, which contained information that was beneficial for vocabulary development for ELL students.

Teacher Modeling

Both Amy and Bob used PowerPoint to visually display text and discuss pictures in order to develop student vocabulary. Amy stated, "The way I teach vocabulary with my children is that I create PowerPoints, so I use my Interactive Whiteboard to teach the words so that they are jumping off the screen" (Amy, interview, March 13, 2014). Displaying vocabulary through pictures allowed the students to directly have a visual to connect with the word. Bob used the visual aspect to display the poem as he read the text out loud. Bob also used the IWB to display the word families on the projector and display the text of the poem that they would be reading together (Figure 4-9). Both teachers used the IWB to model and display the text and pictures.

Implementing Learning Strategies

One learning strategy was displaying visuals and pictures to assist with vocabulary development. Amy felt using the projection aspect to build student vocabulary was effective because the students were able to see the picture and the word to make an association. Having a visual like a picture helped with ELL student vocabulary development by providing students with a picture of what they were actually learning. Through the use of a program like PowerPoint, vocabulary was demonstrated and visually presented to the students (Figure 4-10). Amy presented with PowerPoint to review the words many different times with pictures to make sure that the students understood and made a connection. Amy also referenced using the projection aspect of the IWB to display videos to the students, which assisted with academic content vocabulary in subjects like social studies and science.

Bob also used this strategy of visual representation when he started his lesson by describing what the students would be learning during the lesson and displayed the word families on the IWB. Bob also provided an example of software that he used called Visual Words in which students type in a word and the software shows all of the words and pictures that are associated with it. This software helped with word association, which was, “another technique that is useful especially when teaching a science lesson” (Bob, interview, April 29, 2014). This software could specifically be useful for teaching tier-two and tier-three words. Both teachers felt displaying visuals and pictures was particularly effective in supporting student vocabulary development through providing quality input for students, which was a central component of comprehensible input.

The projection aspect of the IWB was effective with differentiating lessons for students particularly with multiple intelligences. Bob said, “When something is projected

on the whiteboard, it gives access to that which is on the screen" (Bob, interview, April 29, 2014). The IWB lent itself to being very visual. Both teachers were able to display the text, which allowed the teachers to visually show and discuss the words. They were able to model reading the vocabulary, which allowed the teachers to read the words out loud (Figure 4-11). During Amy's lessons, she was able to teach the words in context because the students were not able to acquire this language in the home in a contextualized manner. Being in the UAE, the students were able to use the visual aspect of the IWB to see the vocabulary in different contexts.

Supporting Bi-literacy

Bi-literacy was supported because the students were able to visually see the words in English and Arabic. Although the output was not in Arabic and English during Amy's lesson, she displayed the PowerPoint to the students where she taught the vocabulary in English and Arabic. She was able to ask students about the vocabulary words and stated, "Who knows the Arabic word for _____?" After doing this, the picture was displayed, and students were able to read and see the vocabulary. Amy asked questions about each of the vocabulary words that the students spoke. The students also played a game that encouraged them to use the words in English and Arabic (Figure 4-12). Vocabulary development for ELL students was supported through visual learning with the IWB. The IWB primarily used Explicit Vocabulary Instruction, which matched words directly with their picture, supporting vocabulary development.

Additional Findings

This study explored how the features of the IWB supported vocabulary development; however, there were additional findings for the use of the IWB in the

classroom. Those findings focused on perceived benefits of using IWBs, improvements and challenges, motivation and engagement.

Perceived Benefits of Using IWBs

In addition to the themes discussed above, while observing and working with both teachers, there were multiple perceived benefits from using the IWB. Within Amy's class, the IWB assisted with multiple areas, such as modeling handwriting, letter formation and reading Arabic words. These features are also applicable to other academic areas like mathematics and science. In Amy's class, the students also benefitted from being able to interact with and use the IWB markers. Amy also felt the IWB was "child friendly" and encouraged students to make and correct their own mistakes. According to Amy, "As a result of using the IWB, you are able to meet the learning outcomes in an interactive and engaging way" (Amy, interview, April 1, 2014). Increasing interactivity was one of the perceived benefits of using the IWB.

Another of the perceived benefits was the facilitation of student participation. Both teachers felt using the IWB encouraged students to participate at a higher rate. Bob also spoke about how the IWB facilitates students having success as a class. Bob stated, "Their motivation was still there at the end. They learned the language, they manipulated the language and finally they used the language" (Bob, interview, April 29, 2014). There were several perceived benefits to using the IWB that could be applied to other subjects than ELL and vocabulary development.

Improvements and Challenges

One of the challenges faced by the teachers was the lack of resources and equipment available (Amy, interview, March 13, 2014). In some instances, the equipment was faulty and students had difficulty using items like the IWB markers. For

instance, Bob stated, “Some students still have challenges when you are using that type of marker. Sometimes this caused a delay” (Bob, interview, April 29, 2014; Bob, interview, May 7, 2014). This was one of the improvements that both of the teachers suggested in their interviews. The IWB markers that were used were older and needed to be updated in order to use them properly in future lessons. The height of the whiteboard was also a minor issue that could be addressed through maintenance at the school. Several of the smaller students struggled when attempting to answer questions that were at a physically higher level.

An improvement that both teachers felt could be implemented was allowing students to take ownership of their learning through allowing them to create their own activity. This suggested improvement will have an impact on the future design of the lessons. The teachers also felt the students should be allowed to work on the IWB without teacher input so that they could correct their own mistakes. Although this was implicit in the activity, both teachers felt this could be improved in future lessons.

Motivation and Engagement

Although motivation and engagement are difficult to measure and were not a focus of this study, both teachers felt the students were excited to use the IWB. For instance, Amy stated, “For my boys, it was engaging. They were very engaged because the Interactive Whiteboard lends itself to that” (Amy, interview, March 22, 2014).

The teachers also made note that using the IWB markers was engaging for the students. Amy stated:

> I think that it is good in the sense that is engaging. The children already have the knowledge, so they have the ability to navigate around with the make up the IWB. You do not have to reteach it. It is very child friendly. Not afraid to make mistakes. As a result of it you are able to meet the

> learning outcome that means that you can meet the outcomes in an interactive and engaging way" (Amy, interview, April 1, 2014)

From the observations, interviews and reflections of the students in both classes, they appeared to enjoy the lesson and felt successful. They were able to, "access the technology and show their other classmates what they are doing" (Bob, interview, April 29, 2014). Students were also engaged in games and competition in both lessons, which assisted with motivation. The teachers reported student participation and interaction was high in both lessons.

Summary

Results and findings from the data collection and analysis helped identify uses of the IWB for ELL students in classrooms in Abu Dhabi, United Arab Emirates. The findings demonstrated various ways that ELL student vocabulary development can be assisted through the implementation of IWB lessons. Chapter 5 contains a discussion of the implications for future work and research.

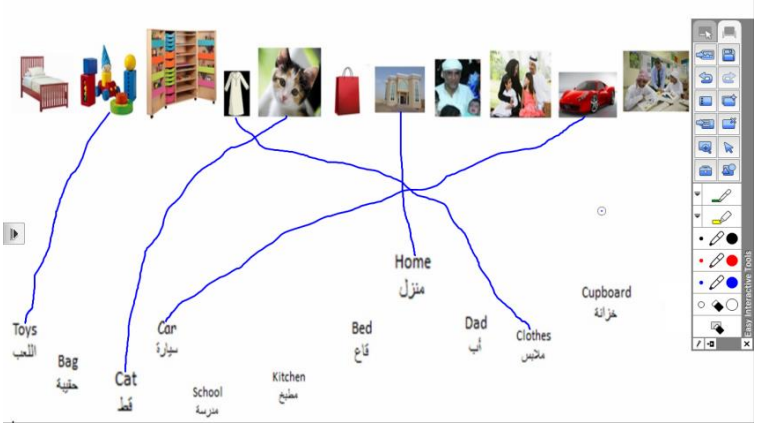

Figure 4-1. Using the marker for reviewing vocabulary.

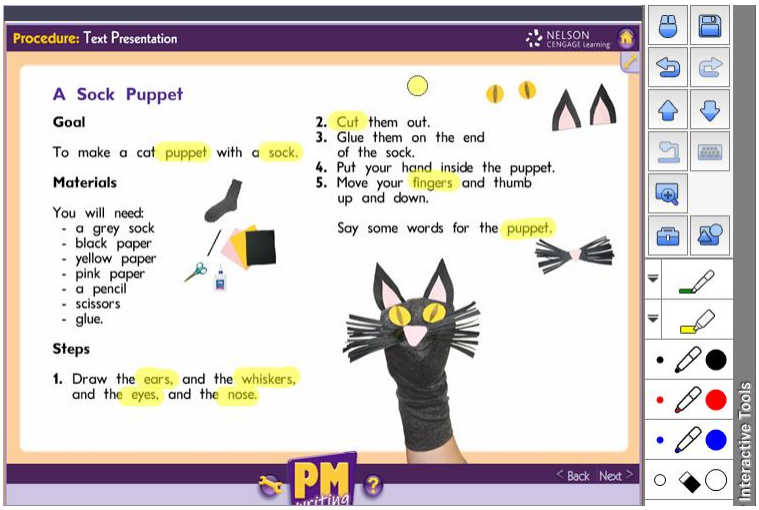

Figure 4-2. Utilizing the highlighting tool of the IWB markers.

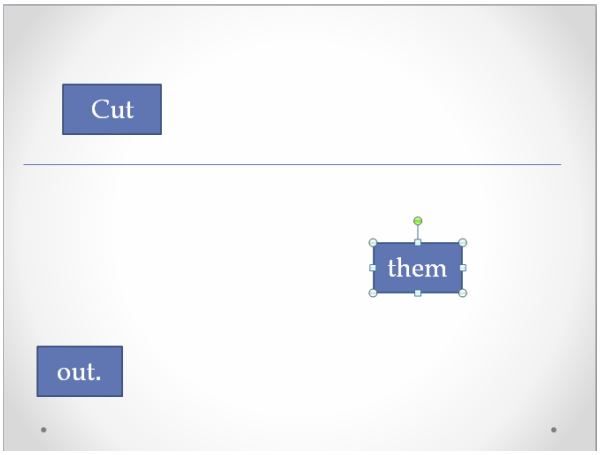

Figure 4-3. Utilizing the drag and drop tool for simple sentences.

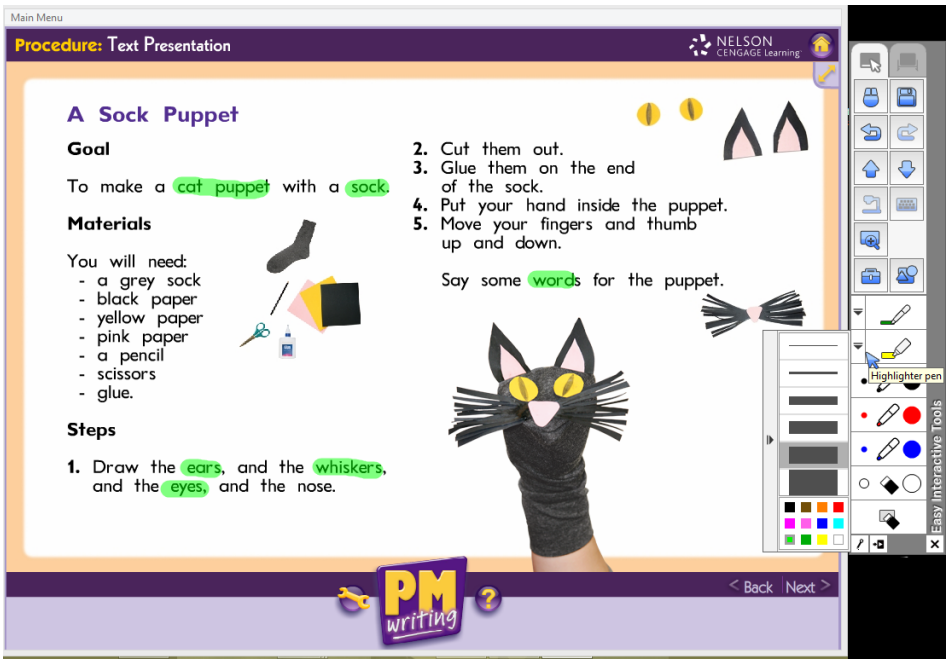

Figure 4-4. “A Sock Puppet” Read aloud with highlighting feature

Figure 4-5. Screenshot of placing shapes around the word families.

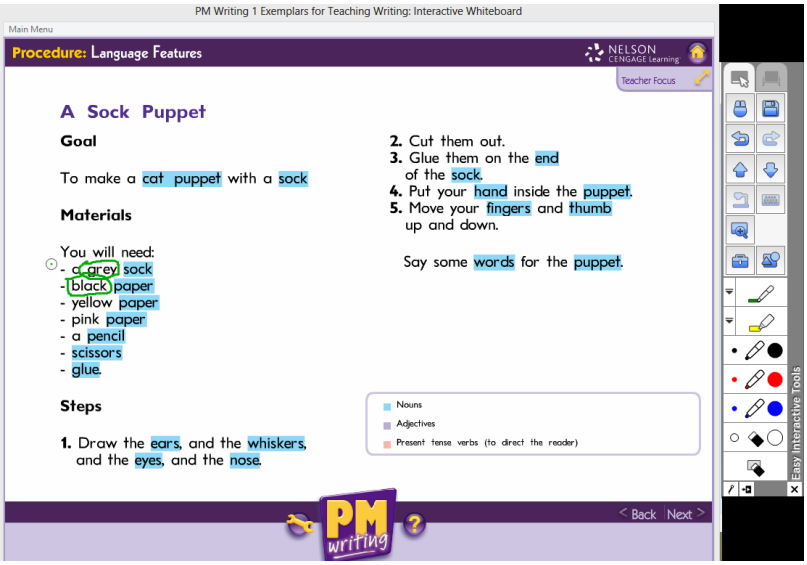

Figure 4-6. “A Sock Puppet” with text highlighting features

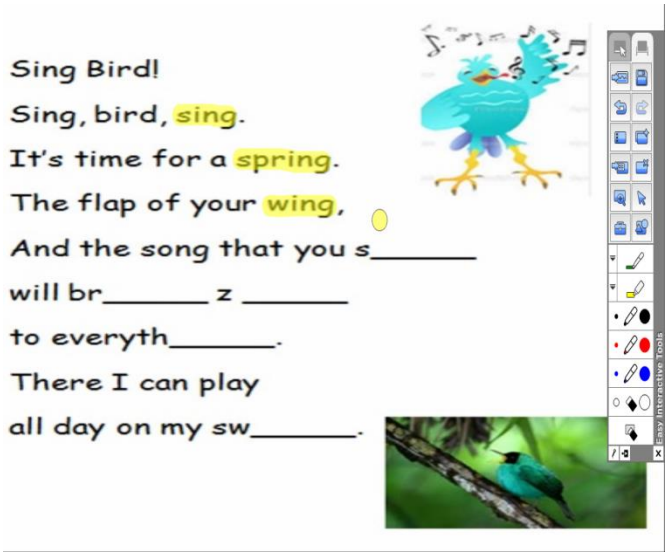

Figure 4-7. Highlighting specific word families

Figure 4-8. PowerPoint presentation for “Creating a Sock Puppet”.

Figure 4-9. “Sing Bird!” poem text

Figure 4-10. PowerPoint slides in English and Arabic

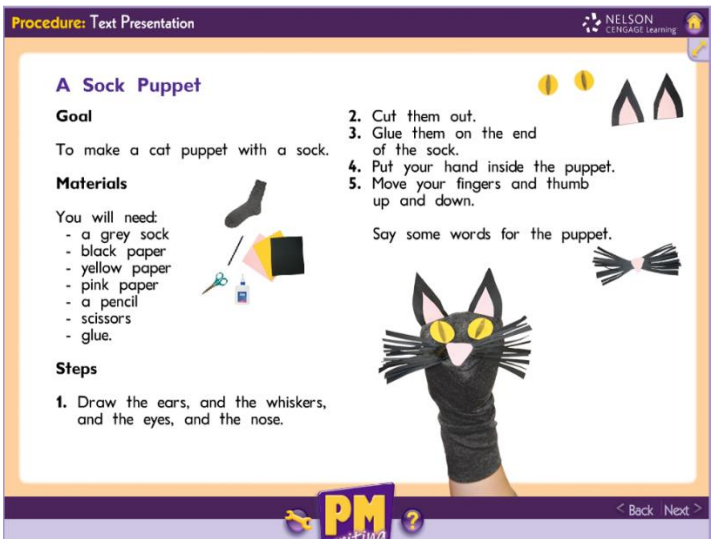

Figure 4-11. “A Sock Puppet” text for the read aloud

Figure 4-12. Practice game for the words in English and Arabic

CHAPTER 5
DISCUSSION AND IMPLICATIONS

The final chapter of the study includes a summary of the study, a discussion of the findings and implications. I will also make suggestions for future research.

Summary of the Study

In primary schools in Abu Dhabi, teachers are in need of IWB-based lessons and activities. This is because of the availability of technology as well as directives from the Abu Dhabi Education Council (ADEC). Native English-speaking teachers were recruited to work with the local population, which, according to testing scores as well as research from the Abu Dhabi Council, struggles with English language development, particularly vocabulary (ADEC, 2013). However, the implementation of IWB materials to facilitate vocabulary development has not been adequately researched. The purpose of this study was to research how the use of an IWB can assist with vocabulary development for ELL students in a boys' school in the UAE. It is intended to help teachers and instructional designers implement activities utilizing IWBs for ELL students.

The primary research question for this study was: how can IWBs be used to facilitate vocabulary development in an ELL classroom?

I first created two activities based on Second Language Acquisition theories (SLA) (Ellis, 1998; Krashen, 1981; Short, Echevarria, & Richards-Tutor, 2011) and principles for using an IWB (Hur & Suh, 2012; Interactive Technologies in Language Teaching, 2011; Wall, Higgins, & Smith, 2005; Wood & Ashfield, 2008). These lessons were implemented by two teachers, who also provided input on the lessons before each implementation, in four classrooms. Amy provided input on the oral language portion of the lessons. Bob gave feedback on the differentiation aspect of the lessons. When

designing lessons and activities for ELL students, the teacher or designer must attempt to make language comprehensible through the use of modified language input, visuals and graphic organizers (Krashen, 1981). Additionally, if the lessons integrating IWB activities are not designed by the teacher, teacher input or adaptation of the lesson to a specific classroom or context is essential.

To determine how these lessons contributed to student vocabulary development, I utilized an exploratory, qualitative research method design for data collection and data analysis. I observed each teacher twice for a total of four observations. Before and after each observation, I conducted pre- and post-interviews. During the interviews and in between the lessons, we made changes to the activities based on vocabulary development and teachers' feedback. The lessons were observed using an adaptation of the Sheltered Instruction Observation Protocol (SIOP) Model and International Society for Technology in Education's (ISTE) classroom observation tool (ICOT) (Echevarria et al., 2000; Fledderus et al., 2011; ISTE, 2013). The teacher interview questions were designed to understand teachers' perceptions of the usage of the IWB for vocabulary development of their students. The interviews and observations were recorded, and I maintained a reflection journal throughout the study.

Discussion of Results

There are three major findings from this study on using IWBs in ELL classrooms: Benefits of teaching with IWBs, multimodal aspects of IWBs and vocabulary development. The research question focused on, "How can IWBs be used to facilitate vocabulary development in an ELL classroom?" The data from the participants indicated there were some benefits to using the IWB in classrooms. The data supported the literature on comprehensible input in having content that is appropriate to (slightly

higher than) the student's English proficiency level (Echevarria, Short, & Vogt, 2008). The IWB-based lessons gave students the opportunity to correct their mistakes and to clarify concepts related to vocabulary (Wu, 2010). Giving students the opportunity to arrange simple and complex sentences as well as participate in games related to the vocabulary supports this aspect of the literature.

Multimodal Aspects of Using IWB

The multimodal aspects of the IWB that emerged were tactile, audio and visual, which was supported from the literature (Chapman, 2005; Cuban, 2001; Glover & Miller, 2002). Many students benefit from different learning styles that help students to acquire vocabulary (Wu, 2010). Effective usage of the IWB depends on being informed about the principles of instructional design, and implementation should incorporate the multimodal aspects of the IWB. Technological tools like the IWB can help second-language learners to develop through providing multiple ways to access information through comprehensible input (Echevarria, Short & Vogt, 2008; Ellis, 1998; Wu, 2010). This technology can be used to supplement instruction in the classroom, which must take into consideration the use of modified language input for ELL students (Echevarria, Short, & Vogt, 2008; Krashen, 1981; Beck et al., 2002). The use of multiple modalities at the same time differentiates the IWB from regular teaching techniques. Using the IWB markers with the lesson increases interactivity with digital items in a way that regular teaching does not. Utilizing the visual and audio aspects of the IWB can be similar to using printed pictures or audio CDs in the classroom. The difference is that these materials can easily be implemented seamlessly into lessons through the use of digital materials.

Vocabulary Development

Literature on vocabulary and second-language acquisition states that instruction for second-language learners should be focused on meaning rather than grammar (Krashen & Terrell, 1983; Valdez, 2012). This study utilized IWB-based lessons in order to help students to acquire basic interpersonal communicative skills (BICS) at the tier-one word level (Beck et al., 2002; Cummins, 2002). The results showed the IWB-based lessons were effective in assisting students with learning select 'first-tier words' out of the three tiers (Beck et al., 2002). The methods and techniques used during the lessons have potential to help students acquire vocabulary (Asbeck, 2012). The techniques used in this study can also assist with acquired Basic Interpersonal Communicative Skills (BICS) (Cummins, 2002). This study was conducted at the elementary level, and the techniques used could be applied to second-language learners at secondary and tertiary levels.

Several ways to develop vocabulary came forward during the lessons. Pre-teaching vocabulary, differentiating instruction, supporting bi-literacy, teacher modeling, implementing learning strategies, using various resources, utilizing repetition and using grouping and assessment strategies were techniques that emerged for vocabulary development. Both teachers pre-taught the vocabulary through various techniques like displaying the words on the IWB. Instruction for students was also differentiated by having higher-level work and having the students think about their learning rather than just doing.

In each of the classrooms and lessons, students' vocabularies were developed through the focus on bi-literacy and having the words in English and Arabic (Krashen & Terrell, 1983). The teachers were also able to model proper vocabulary usage for

students through the use of the IWB. Using the IWB markers to model vocabulary for teachers helped to present the information in a way that students can comprehend (Beck et al., 2002). For instance, in one lesson, the teacher was able to model oral language usage, which gave the students the ability to make mistakes without pressure. The teachers also pointed out several learning strategies that they were able to implement. One focused on teaching vocabulary through decontextualized teaching, which allowed the teacher to focus on the specific vocabulary being taught. In another instance, the teacher was able to help the students to order and use the words in simple and complex sentences.

Repetition of vocabulary was one area that came up many times. The students were able to hear the vocabulary in English and Arabic multiple times. In the interviews, both teachers felt repetition was important for ELL student vocabulary development. The students were given the opportunity to practice with the words (Graves, 2006). The teachers also felt grouping and assessment strategies were beneficial both for their data collection and for the students. They felt formative assessment could help them when designing further lessons. The literature supports having pictures and words together, which assists with student assessment (Chun & Plass, 1999; Interactive Technologies in Language Teaching, 2011). There was evidence from the data analysis and findings to suggest that the IWB-based lessons were effective in assisting the ELL students with vocabulary development. The tactile, audio and visual aspects of the IWB helped to make language comprehensible, which assisted with vocabulary development (Interactive Technologies in Language Teaching, 2011; Wall, Higgins, & Smith, 2005).

IWBs can also be beneficial for teaching high level content and academic language. IWBs can assist with this by providing relevance and authenticity during language learning (Maher, 2011). IWBs help to demonstrate material through the enabling of student and teacher discussion which can assist with learning higher level academic topics (Jang & Tsai, 2012).

Benefits of Teaching with IWBs

Analysis of the data revealed multiple features of the IWB that are supported by the literature. Several benefits of the IWB were utilizing flexibility and versatility, supporting planning and the development of resources and increasing interactivity and participation in lessons (Levy, 2002; Wall et al., 2005). These benefits are applicable to other academic subjects like mathematics and science. IWB usage provides the ability to manipulate vocabulary through the integration of digital media (Hall & Higgins, 2005; Schmid, 2008). Students were able to practice their reading and vocabulary skills using a drag and drop activity where they had to create multiple types of sentences. Interactivity was increased through using words and images in both lessons. Increasing use of visual aids is a key component of comprehensible input and second-language acquisition. Lessons were centered on the learner and focused on the authenticity of the vocabulary (Schmid, 2010). Students were also required to read and highlight vocabulary, which utilized the tactile aspect of the IWB. Audio use was also used in each of the lessons. The IWB was utilized to display and hear vocabulary in English and Arabic. Visuals were also a major feature of the IWB. PowerPoint was used in two ways: to display visuals and to allow students to create different styles of sentences. Pictures were also important for word association, where visuals were also used to practice reading, listening and speaking. In addition to second-language acquisition,

IWBs were also useful for motivation and engagement (Gerald & Widener, 1999; Knowlton, 2008; Tanner & Jones, 2007).

Affordances of Interactive Whiteboards

There a number of affordances that the IWB provides for ELL students. Teaching is supported through the integration of activities where ELL students can interact with the board either by utilizing the drag and drop feature or responding to discussion centered on the material that is shown on the board (Teck, 2013). Incorporating formative and summative assessment with the IWB is a way to help students with exploration, explanation and elaboration.

IWBs also can contribute utilizing plenaries to review instruction. For instance students can present their work by scanning or projecting photos onto the board for discussion. The IWB facilitates reflection, commenting, and correction by the class and teacher. These methods of instruction are important for ELL students to understand how to properly use the English language. Having authentic tasks can provide real world relevance for ELL students. For instance IWBs can allow students to utilize online news sites where teachers can facilitate discussion. IWBs also help teachers to easily incorporate resources such as Google Earth and You Tube which can also help to facilitate discussion (Hur & Suh, 2012; Slay, Sieborger & Hodgkinson-Williams, 2008).

IWBs can facilitate the control of the computer with the touch of a screen. Teachers can model language and vocabulary ELL students. Teachers are able to pay more attention to the class rather than to the computer screen which is what this affordance allows. The capture feature of the IWB allows teachers to review and revisit lessons for reinforcement. This affordance also allows teachers to reflect on their practice.

IWBs can have a positive effect on student learning and help students with social interactions with hands on tasks (Hur & Suh, 2012). IWBs also incorporate various multimedia and collaborative projects (Slay, Sieborger & Hodgkinson-Williams, 2008). The encouragement of critical thinking when completing group projects as well as encouraging students to be active participants in learning (Mott, Sumrall, Rutherford, Sumrall & Vails, 2010).

IWBs assist with second language learners by using animations which can provide relevance and authenticity during learning (Maher, 2011). By utilizing this function of the IWB this encourages students to engage with the material. IWBs can improve whole class teaching along with student discussion which can aid with helping teaching to explain complex academic topics (Jang & Tsai, 2012). This allows teachers to demonstrate the material in a way which reveals students' development with the material such as vocabulary.

One of the areas that could be incorporated is through having students to have open ended activities with the IWB rather than practice that is authoritatively guided by the teacher. An affordance that I did not include in the lesson design was the inclusion of videos or animations. Students were also not given a collaborative or group project. These are some affordances that can be included in future lesson planning.

The IWB and Technology Integration

The Technology Integration Matrix (TIM) (Florida Center for Instructional Technology, 2014; Jonassen, Howland, Moore, & Marra, 2003) characterizes meaningful learning environments with technology integration as: active, collaborative, constructive, authentic and goal-directed. The levels of technology integration into the curriculum are: entry, adoption, adaptation, infusion and transformation. The focus of

this study was not to measure technology integration in the four ELL classrooms according to the TIM, but the having the TIM provide a valuable lens to discuss how the IWB activities were integrated in the classrooms in this study. The usage of IWBs in this study did not correspond to every single level in the TIM, but there was some overlap in the way that the IWB activities were used by teachers in ELL classrooms.

The first characteristic of creating a meaningful learning environment is that the students are actively using the technology in the classroom. Each of the activities that the students used with IWBs allowed the students to conventionally and independently use the technology. The highest level, which focuses on unconventional use of tools, could be improved in future lesson design of IWB activities. The use of games and competitive activities with the IWB was successful and allowed the students to work collaboratively. Some of the activities could have been completed without the technology; however, the games and manipulative aspect of IWBs allowed students to interact with the IWB and each other and build their knowledge. Such manipulatives also facilitate second-language acquisition (Krashen, 1981), and other researchers have reported that IWBs can be beneficial for enhancing interactivity within classrooms (Ansell & Park, 2003; Gerald, Greene, & Widener, 1999; Tanner & Jones, 2007). During the lessons, learners were able to learn constructively through the use of activities and games.

The learning and use of technology was authentic because of the additional activities that were provided for students. While some higher-order-thinking activities were included to have students build their own sentences using the vocabulary, this area could be improved in future lessons, which was also suggested by the teachers in

this study. The vocabulary that was taught were items that would be used on an everyday basis, thus building on authenticity and relevance of the topics being taught. The images of common vocabulary were presented with Arabic captions; therefore, students had a reason for learning the material being taught. Several of the activities allowed the students to correct their own mistakes, which is a tenet of goal-directed learning.

Motivation and Engagement

Prior research has found that IWBs provide the opportunity to increase levels of attention and engage students (Gerald & Widener, 1999; Knowlton, 2008; Tanner & Jones, 2007). IWBs have also been proven effective when addressing student motivation (Beeland, 2002; Lewin et al., 2008). Although not a focus of this study, in the interviews, teachers mentioned students' high motivation during the lessons due to the use of the IWB, and this was also apparent in my observations. The students were very engaged during the lessons, which helped with getting students interested in the lesson content. Student participation was high particularly during the vocabulary games that were facilitated by the IWB. Students were engaged in competition, which helped to lower their affective filter. They were able to feel comfortable in the classroom setting, which enabled them to communicate in a natural manner. When students feel comfortable in their environment, then second languages are acquired more efficiently (Krashen & Terrell, 1983). Second language acquisition theory makes the argument for the positive benefit of motivation and engagement for second-language acquisition. Both teachers reported student participation and interaction was high in all the lessons. Although participation, interaction and motivation were not measured, the data collected in this study suggest that incorporating games and competition into IWB-based lessons

can help facilitate vocabulary development. This is an area for future research using an experimental design.

Implications

Based on the results of my study, this section first presents the implications of using IWBs in my practice and for working with teachers. It then discusses the implications for designing-IWB based lessons and the implications for schools and professional development for teachers and administrators.

Implications for My Practice

When I first started thinking about designing this study, I was working as a second-grade teacher with the Abu Dhabi Education Council. I had taught for three years in the United States and one year in the United Arab Emirates, and though I had used IWBs in the United States, I did not have experience designing and implementing IWB-based lessons for first- or second-language learners in an academic setting. As I have collaborated with teachers and designed IWB-based lessons, several implications have come forward for my instructional practice and for other teachers. First, there are several benefits of implementing not only IWBs but also other types of educational technology in classrooms in the UAE and around the world (Interactive Technologies in Language Teaching, 2011). Second, I have started to implement second-language acquisition principles like comprehensible input and have applied research on instructional techniques for ELL learners in my classroom (Echevarria, Short, & Vogt, 2008). Lastly, many of the techniques and lessons learned in this study can be presented for teachers through professional development.

The IWB increased interactivity and participation in lessons, according to the study. The principles applied to IWBs in assisting with motivation can be applied to

other educational technology applications, such as Web 2.0 tools and educational computer programs. After implementing this study, I have more of a focus on student-learning styles in my practice than I had before. Previously, I would just place the activity in front of the students without thinking about how they best learn. Moving forward, I have started to think about how to best to implement educational technology in the classroom through focusing on the multimodal aspects of each technology-based activity. Teacher modeling of technology was important, as evidenced from the observations and feedback from teachers. In my own practice, I have made sure to consistently model how to properly use educational technology applications in my classroom. Students should also use educational technology to create and manage their own learning rather than just doing because this increases ownership and engagement in learning. While completing teacher-designed activities is a first step for students to learn, student-created examples or activities can help students practice their knowledge. When thinking about creating and designing lessons using educational technology, teachers should be cognizant of higher-level students and vary activities. After conducting this study, in my own classroom I have started to try to create activities that allow students to create at a higher level. In this instance, my practice has been impacted.

By implementing this study, I have learned much about second-language acquisition theories and principles. Although I am not teaching in the UAE anymore, I feel implementing second-language acquisition practices have relevance to students in the United States, not only ELL students. For instance, second-language acquisition principles focus on having high levels of target language input, the use of visuals and

opportunities for discussion and concept clarification. I am currently teaching third grade in the United States and have students who do not perform as well in academic subject areas. For instance, I recently taught two social studies units on Mary McCleod Bethune and Paul Revere. I made sure all of my activities included many visuals and any opportunities to discuss both of their lives and their importance. Much of the vocabulary that we use in social studies and science lessons are beyond the students' comprehension levels; therefore, lessons and activities based around comprehensible input are applicable to teaching students at these levels. Lessons that I teach native English speakers utilize many of the second-language acquisition principles and techniques that I implemented in the United Arab Emirates. Many of these techniques for educational technology and second-language acquisition can be utilized by teachers in the United Arab Emirates and around the world.

Implications for Instructional Design

Previously, I had experience utilizing Gagne's nine events of instruction to design lessons during my qualifying project (Gagne, 2005; Martin, 2012). This study utilized some of those principles but included more second-language acquisition principles and was based on the SIOP model, which incorporated the following: learning standards, key vocabulary, materials, main lesson, guided/small group instruction, individual practice/assessment and a closing (Short, Echevarria, & Richards-Tutor, 2011). The lesson design was inclusive of multiple modalities, whole class teaching techniques and small-group instruction (Coyle, Yanez, & Verdu, 2010; Levy, 2002). The combination of instructional design principles and theories specific to the discipline or subject being taught can enhance the value of a lesson design and can effectively teach a topic using technology.

There are other implications for the instructional design of the lessons/materials and software in order to effectively teach students. The main lesson and closing as well as other portions of the lesson plan were important to effectively implement the lessons in the ELL classrooms in this study. One area that could be changed would be to include some overlap between the specific parts of the lesson plan. For instance, guided/small-group instruction tended to overlap with individual practice/assessment; however, this occurred organically during the lesson and was not written into the lesson plan and design. The implementation of the lessons indicated that one area that could be improved is the addition of a differentiation/student ownership section. Both teachers mentioned this in their interviews and suggestions for improvement. Both lessons utilized a modified version of the SIOP model or followed some of the second-language principles in order to create and design lessons. Based off of the observations and interviews, this model was effective. These parts of the lesson are not only applicable to student instruction at the elementary level but can be applied to middle, secondary and tertiary schools. Academic instruction can utilize many of these areas when designing instruction in areas such as science, mathematics and reading.

This study utilized several pieces of educational technology software with the IWB. There were several implications gained from using this software, such as Easy Interactive Tools©, PM Writing IWB software©, Microsoft PowerPoint©, Memory Game Maker© and Sentence Game Maker©. Design of activities with any software should strive to incorporate multimodal features. For instance, when using Easy Interactive Tools©, the PM Writing IWB© software and Microsoft PowerPoint©, this study incorporated visual input, tactile input and sound/auditory learning. Visual input was

utilized through the use of pictures and images. Sound/auditory learning was included through the use of pre-recorded language in English and Arabic. Easy Interactive Tools© allowed students to interact with the vocabulary through tactile input like highlighting or circling. Though it can be difficult to incorporate these three learning styles at the same time, instructional designers should attempt to include each when designing activities or lessons. These learning modalities are not only applicable to second-language learners, but they can be applied to lessons catered for native English speakers. Another area that should be addressed is the provision of more opportunities to use the IWB markers in each area of the lesson. Students were very excited about using the IWB markers and interacting with the board, so a suggestion based on this study is to provide sufficient time for tactile interaction. The students were also excited about using Memory Game Maker© and Sentence Game Maker©, which allowed the students to participate in a competitive game. More opportunity could be given for them to compete against each other in vocabulary-based activities. Both the inclusion of games and interactions with the IWB can motivate students and increase their willingness to learn.

Implications for Schools

During the study, both teachers had some experience with using IWBs; however, they needed a bit of training using the software – both the one that I installed, like Memory Game Maker©, and pre-loaded software, such as Easy Interactive Tools©. The literature suggested teachers who had more than two years of experience using the IWB should utilize each of the features effectively (Levy, 2002; Wall et al., 2005). Teachers in the UAE environment come from many different backgrounds. Some may have experience with educational technology and second-language acquisition,

whereas some may have no experience at all. Even within this study, both teachers had different backgrounds and different ideas on how to effectively use IWBs within their classrooms and how best to teach vocabulary to their students. Teachers should be provided guidance on how to find effective lessons, activities and software. There is a need for a standardized set of professional development for teachers and administrators in Abu Dhabi. Such training can also be applied to other countries in the Middle East.

Professional development implementation should be focused first on the individual teacher. Bob had more experience using IWBs than Amy. He would require different content in terms of professional development than she would. Teachers and administrators in the Abu Dhabi School system are from varied countries, such as the United Arab Emirates, New Zealand, the United States, South Africa, Australia, Canada and the United Kingdom. Given the diversity of experiences, in addition to educational technology professional development that is foundational, professional development should also be individualized. In this particular environment, with students and teachers from many different countries, there also is a need for culturally relevant training. In order to support the teachers, the administrators must be aware of how to effectively teach with IWBs and the ways that IWBs can best facilitate teaching and learning through the medium of English. One of the goals of the Abu Dhabi Education Council is to develop and support students' bi-literacy. Most of the teachers do not speak Arabic; therefore, teachers need to have access to the vocabulary in the students' native languages in order to effectively teach during lessons.

Professional development should be focused primarily on tactile uses of the IWB. This study showed several tactile uses with the IWB markers, such as increasing

interactivity with words and images and manipulating words and pictures. Although these uses were applied to second-language acquisition, there are also uses for academic content areas. For instance, in my current teaching position, I teach an early intervention program mathematics course. I recently taught a unit on geometry where students were given the opportunity to manipulate geometric figures like pentagons and octagons through usage of the IWB markers. Teachers can take many of these uses and apply them to areas like science and language arts. Another area of development for administrators is thinking about proper areas to install IWBs in classrooms, an area that came up both in literature and the interviews (Levy, 2002).

Future Research

There are a limited number of studies on vocabulary development for elementary students in the United Arab Emirates. The majority of research on IWBs is focused on the secondary and tertiary levels. This study was designed to investigate IWB activities and vocabulary development of elementary male students in the UAE based on research on IWBs in other parts of the world and second language acquisition theory. There are several areas of research that would make for interesting future research opportunities. Beyond further research at the elementary level for vocabulary development both in the implementation of IWB lessons and the usefulness of certain types of lesson design, there could also be more action research by teacher-practitioners in this specific environment. Given the availability of technology in the classrooms, more research is needed on using various types of educational technology in the United Arab Emirates. Teacher and administrator professional development in implementing educational technology in Abu Dhabi is an area that also should be further researched and studied.

Conclusion

In conclusion, this study presented several multimodal features (Interactive Technologies in Language Teaching, 2011; Wall, Higgins, & Smith, 2005) of using IWBs for vocabulary development in ELL classes in Abu Dhabi, United Arab Emirates. The results of the study support the literature on second-language acquisition, vocabulary development and benefits to using IWBs in classrooms. The data presented areas of usage for developing vocabulary, IWB features, motivation and engagement. The data indicated areas of improvement and suggestions. There are also implications for my teaching practice, instructional design and schools.

APPENDIX A
TEACHER INTERVIEW PROTOCOLS

Pre-Interview Protocol

Interviewee (Title and Name): _____

Interviewer: _____

Date: _____

Thank you for signing on to work on this research project. This research project focuses on the use and design of lessons for an interactive whiteboard with an interest in vocabulary development. This study does not intend to evaluate your teaching or techniques but rather to learn about how to improve techniques for teaching through the use of these materials. I have planned this interview to last no longer than 60 minutes. I will record this interview for further transcription.

Background

1. Can you tell me about your background and teaching experience with ESL students in the United Arab Emirates?

Vocabulary Instruction

2. In what ways have you taught vocabulary to your students in the UAE and your native country? What techniques and methods have you used?

Interactive Whiteboards and Technology use

3. How do you feel about using technology/ interactive whiteboards in the classroom?

   How have you used an interactive whiteboard in your teaching practice?

   What benefits and challenges have you experienced when using the interactive whiteboard in the past?

4. What other types of technology do you use in your class?

Feedback on the lesson

5. What feedback do you have about the interactive whiteboard activities for this study?

a. Mini lesson.
b. Small group.
c. Closing.
d. Overall vocabulary instruction.

6. Are there any changes that you would like to make before implementing the lesson in your classroom?

Post First Observation Interview Protocol

Interviewee (Title and Name): _____

Interviewer: _____

Date:_____

I would like to speak with you today to go over the lesson which I observed. Everything that we speak about will be held confidential. I have planned this interview to last no longer than 60 minutes. I will record this interview for further transcription.

This research project focuses on the use and design of lessons for an interactive whiteboard with an interest in vocabulary development. This study does not intend to evaluate your teaching or techniques but rather to learn about how to improve through the use of these materials.

Reflection on the Interactive Whiteboard activities

1. What do you think about how the lessons went?

7. How did you think the interactive whiteboard assisted in vocabulary development in your classroom?

8. Were there any issues or did you face any challenges during these lessons?

Feedback on the classroom observation

9. Here is some of my feedback from the classroom observation. This feedback is not intended to critique your teaching but to strictly look at student interaction with the interactive whiteboard (Guided discussion using the classroom observation tool).

    a. Interactive Whiteboard usage.
    b. Lesson Preparation.
    c. Comprehensible Input.
    d. Strategies.
    e. Interaction.
    f. Practice and Application.

Requesting feedback from the teacher

10. What feedback do you have about the interactive whiteboard activities and lesson?

    a. Mini lesson?

b. Small group?
c. Closing?
d. Overall vocabulary instruction?
e. Overall use of the interactive whiteboard?

11. Are there any changes that you would like to make before implementing the second lesson in your classroom?

Post Second Observation Interview Protocol

Interviewee (Title and Name): _____

Interviewer: _____

Date:_____

Thank you for participating in this study and inviting me in to your classroom. I would like to ask some closing questions. I have planned this interview to last no longer than 60 minutes. I will record this interview for further transcription.

Vocabulary Development:

1. Overall, how would you describe the usage of the interactive whiteboard for vocabulary development of your students? (Probe for areas that assisted vocabulary development and challenges faced.)

Activity Design and Implementation:

2. How would you describe the lesson and activity implementation?

Interactive Whiteboard:

12. After using the whiteboard for these two lessons, do you feel that you will use it in the future? Why or why not? Would you use it differently?

Following Up:

13. Here is some of my feedback from the second classroom observation. This feedback is not intended to critique your teaching but to strictly look at student interaction with the interactive whiteboard (Guided discussion using the classroom observation tool).

    a. Interactive Whiteboard usage.
    b. Lesson Preparation.
    c. Comprehensible Input.
    d. Strategies.
    e. Interaction.
    f. Practice and Application.

Reflection:

14. What changes or modifications would you further make to the lesson?

15. What types of activities would you like to see used with the interactive whiteboard for vocabulary development?

Thank you for allowing me in your classroom and working with me on this study. I would like to stress that all of the information that you provided will be kept confidential. I will share the results of this study with you after finishing the data analysis and writing. Thank you.

APPENDIX B
CLASSROOM OBSERVATION TOOL

1. Setting		
Date:	Observation Start Time/End Time:	# Students:

2. Description of the Room (including placement of the interactive whiteboard) and Characteristics of the Students

2. Description of interactive whiteboard usage, focus on student interaction with the interactive whiteboard and time that was spent using it
Mini lesson:
Small Group Instruction:
Closing:

Written Observation Outcomes

1. How was the interactive whiteboard used for vocabulary development?

2. Recommendations?

	Notes
Lesson Preparation Supplementary materials used to a high degree, making the lesson clear and meaningful (e.g., computer programs, graphs, models, visuals)	
Comprehensible Input A variety of techniques used to make content concepts clear (e.g., modeling, visuals, hands-on activities, demonstrations, gestures, body language)	
Strategies Scaffolding techniques consistently used assisting and supporting student understanding (e.g., think-alouds)	
Interaction Grouping configurations	

support language and content objectives of the lesson	_____ _____
Practice and Application Activities integrate all language skills (i.e., reading, writing, listening, and speaking)	_____ _____ _____

Further Comments:

_____
_____
_____
_____
_____
_____
_____
_____
_____

This tool was adapted from:

Echevarria, J., Short, D., & Vogt, M. (2008). Making content comprehensible for nonnative speakers of English: The SIOP model. *International Journal of Learning, 14*(11), 41-49.

ISTE (2013). ISTE Classroom Observation Tool. Retrieved from http://www.iste.org/learn/research-and-evaluation/icot/.

APPENDIX C
INFORMED CONSENT FORM

You are invited to take place in a doctoral research study on student vocabulary development through the use of an interactive whiteboard. This study will be conducted by Charles Martin, a doctoral candidate at the University of Florida. You are a 3rd grade teacher at Sas Al Nakhl School in the United Arab Emirates and were chosen for this study because of your interest in using the interactive whiteboard. This research study will help to find strategies and activities for implementing interactive whiteboard based lessons in ESL classrooms for teachers. Please review the requirements of this study, and let me know if you have any questions or require clarification before starting this study.

Background Information:
The purpose of this study is to understand how English as a second language (ESL) vocabulary development is assisted using an interactive whiteboard in an English class. The purpose is to help other teachers apply best practice for using an interactive whiteboard for ESL vocabulary teaching.

Procedures:
If you agree to be in this study, you will asked to

1. participate in a 30 minutes presentation detailing this study,

2. provide feedback on lessons, and reflect on lessons,

3. participate in three interviews (pre-, -post first lesson, -post second/third lesson). Each interview will last approximately 30 minutes-one hour.

4. co-plan two lessons with the researcher (Charles Martin) based on the Abu Dhabi Education Council standards, and

5. allow the researcher (Charles Martin) to observe your classroom three times during the study for 60 minutes.

Voluntary Nature of the Study:
Your participation in this study is voluntary. If you join this study and change your mind later you may stop at any time.

Risks and Benefits to Participating in this Study:
There are no risks from participating in this study. There are no direct benefits to you for participating in the study.

Sharing Results of the Study:

All the data from this study will be maintained in a secure place. This study's results will be shared either in summary or as part of a doctoral dissertation. Your name and any data associated with you will be kept confidential.

Compensation:
There will be no compensation for participation in this study.

Confidentiality:
All information that you provide will be kept confidential. Your name will not be included in this study.

Questions and Contacts:
If you have any questions please feel free to contact the researchers (Dr. Swapna Kumar or Charles Martin) by e-mail at lmcmtutors@ufl.edu in regards to concerns and questions. You can also contact them by phone at +971 56 618 7925 or +971 56 375 1200. For information regarding your rights as a research participant contact the IRB at 352-392-0433.

Statement of Consent:
I have read the above information. I have received answers to any questions that I have at this time. I am 18 years of age or older, and I consent to participate in the study. Your identity will be kept confidential to the extent provided by law. You may withdraw your consent at any time without penalty.
Printed Name of the Participant

_____

Signature of the Participant

_____

Signature of the Researcher

_____

APPENDIX D
TABLES AND FIGURES

Table D-1. Software/Activities and matching SLA/Educational Technology Theories

Software/Activity	Description of theory and matching references
Easy Interactive Tools©	Learning the meaning of new words (Beck et al., 2002, Stahl & Nagy, 2006)
	Presentation of word families (Edelen-Smith, 1997; Scheiner, 2005)
	Visual Input (Vidot, 2011)
	Tactile Input (Ansell & Park, 2003)
	Modification of Input (Krashen, 1983)
PM Writing© IWB Software	Learning the meaning of new words (Beck et al., 2002, Stahl & Nagy, 2006)
	Associating words with images (Chun & Plass, 1999; Interactive Technologies in Language Teaching, 2011)
	Visual Input (Echevarria, Short, & Vogt, 2008)
	Tactile Input (Hall & Higgins, 2005; Schmid, 2008)
	Appropriate speech, use of visuals, and opportunities for discussion and concept clarification (Echevarria, Short, & Vogt, 2008)
Microsoft PowerPoint©	Associating words with images (Chun & Plass, 1999; Interactive Technologies in Language Teaching, 2011)
	Presentation of word families (Edelen-Smith, 1997; Scheiner, 2005)
	Visual Input (McGregor, Sheng, & Ball, 2007)
	Sound/Auditory Learning

Table D-1. Continued

	(Chapman, 2005; Glover & Miller, 2002)
	Quality input (Snow et al., 1981)
Memory Game Maker©	Associating words with images (Chun & Plass, 1999; Interactive Technologies in Language Teaching, 2011)
	Visual Input (McGregor, Sheng, & Ball, 2007)
	Tactile Input (Hall & Higgins, 2005; Walker, 2003)
	Sound/Auditory Learning (Plass et al., 2003)
	Concept clarification (Echevarria, Short, & Vogt, 2008)
Sentence Game Maker©	Presentation of word families (Edelen-Smith, 1997; Scheiner, 2005)
	Tactile Input (Hall & Higgins, 2005; Walker, 2003)

Table D-2. Study Implementation Schedule

Week	Summary
1	30 minute presentation giving an overview of the study with Amy and Bob Pre- Observation Interview with Amy Changes made to the activities and lessons based on reflection and feedback from Amy
2, 3	First in class observation of Class A-1 Reflect on my own notes and preparation for the first interview. Choose recordings from the interactive whiteboard and excerpts
4	Post- First Observation Interview with Amy Changes recommended and the activities will be discussed Re-plan the next lesson
5	Second in class observation of Class A-2 Reflect on my own notes and preparation for the second interview Choose recordings from the interactive whiteboard and excerpts
6	Post Second Observation Interview with Amy
7	Pre- Observation Interview with Bob Changes made to the activities and lessons based on reflection and feedback from Bob
8, 9	First in class observation of Class B-1 Reflect on my own notes and preparation for the first interview Choose recordings from the interactive whiteboard and excerpts
10	Feedback and Post- First Observation Interview with Bob Changes recommended and the lesson will be discussed
11	Second in class observation of B-2 Reflect on my own notes and preparation for the second interview Choose recordings from the interactive whiteboard and excerpts
12	Post Second Observation Interview with Bob
13, 14, 15	Interview transcription, data analysis, coding, clarification from participants where needed (data checking)
16, 17, 18	Write up findings

Table D-3. Developing Vocabulary

Codes	Sub Codes
Pre- teaching Vocabulary	Pre- teaching vocabulary through the use of the IWB and visuals Pre- teaching the word families through displaying them on the IWB Pre- teaching the word families by reading the text of the poem Pre- teaching the vocabulary through the use of PowerPoint
Differentiating Instruction	Differentiating higher level work for students Differentiating through thinking about learning rather than doing
Supporting Bi- literacy	Supporting Bi- literacy through having the IWB display and hear vocabulary in English and Arabic Supporting Bi- literacy through having students pick their own words in English and translate to Arabic Supporting Bi- literacy through student correction of Arabic in Emirati style Arabic
Teacher Modeling	Teacher Modeling through using the teacher as a monitor Teacher Modeling through giving students the ability to make mistakes without pressure Teacher Modeling of reading a poem Teacher Modeling of how to write words with the IWB marker Teacher Modeling of prediction skills Teacher Modeling of oral language Teacher Modeling of reading vocabulary
Implementing Learning Strategies	Focusing on meaning and understanding Teaching vocabulary through decontextualized teaching Question and response Delayed speech Explicitly stating direction Ordering words in simple and complex sentences Chunking vocabulary Word Association

Table D-3. Continued

Using Resources	Using resources and storytelling to teach vocabulary Using resources such as computer games Using resources such as the availability of a pullout support system Using resources such as different software programs
Repetition	Repetition of vocabulary through a rote learning system Repetition of continuously hearing the word Repetition of vocabulary in English and Arabic Repetition of the vocabulary multiple times through seeing, reading, hearing and saying Repetition facilitated with the IWB Repetition of writing Repetition in identifying words
Grouping and Assessment Strategies	Small group instruction Grouping multiple abilities Assessment

Table D-4. Interactive Whiteboard Features

Codes	Sub Codes
Tactile uses of the IWB	Practicing reading and vocabulary skills with a drag and drop activity Increasing interactivity with words and images Manipulating different words and pictures Identifying word families using the IWB markers Increasing interactivity with the IWB Highlighting and reading vocabulary
Using Audio with the IWB	Using oral techniques to build vocabulary Using the IWB for sound creation and oral language practice Using the IWB to display and hear vocabulary in English and Arabic Giving students practice with oral language
Using Visuals with the IWB	Teaching vocabulary through the use of visuals with PowerPoint Using video and visuals to support vocabulary development Using pictures to make word associations Using PowerPoint to create different styles of sentences Using visuals to practice reading, listening, and speaking
Resources and Training	Need for tactile learning facilitated by technology Need for resources and additional training Additional culturally relevant training
IWB Placement	IWB Placement for student grouping formations IWB Placement for student learning IWB Placement for facilitating learning
Improvements and Challenges	Improving the lesson through using multiple pens Improving the lesson through better placement of the board Improving the lesson through raising the difficulty level for students Improving the lesson through providing multiple shapes for practice Challenges when the markers were not fluid Challenges with the lesson when there was a delay with the IWB marker

Table D-5. Motivation and Engagement

Codes	Sub Codes
Student Ownership	Allowing students to take ownership over their learning Students internalizing their learning Using the IWB gives the students a feeling of success
Preparation	Making sure that students are on task Ensuring that the IWB functions smoothly
Competition	Using the IWB for competition Competition and games as an aspect of vocabulary development Utilizing competition to lower the affective filter
Lowering Affective Filter	Communicating in a natural setting Feeling comfortable in the classroom setting
Motivation and Engagement	Increasing student engagement in the classroom Motivation and participation was high
Classroom Management	Using the IWB for classroom management
Teaching in a Multicultural Environment	Teaching vocabulary in a multicultural environment Availability of culturally relevant resources Having different styles of English accents Making generalizations about how boys learn Experience teaching different ages of children

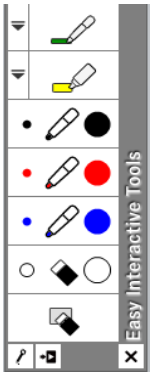

Figure D-1. Easy Interactive Tools© annotation mode features (highlighter, pen, and eraser buttons)

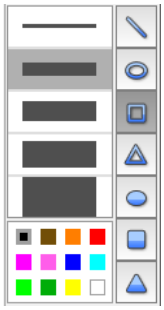

Figure D-2. Easy Interactive Tools© annotation mode features (shape button)

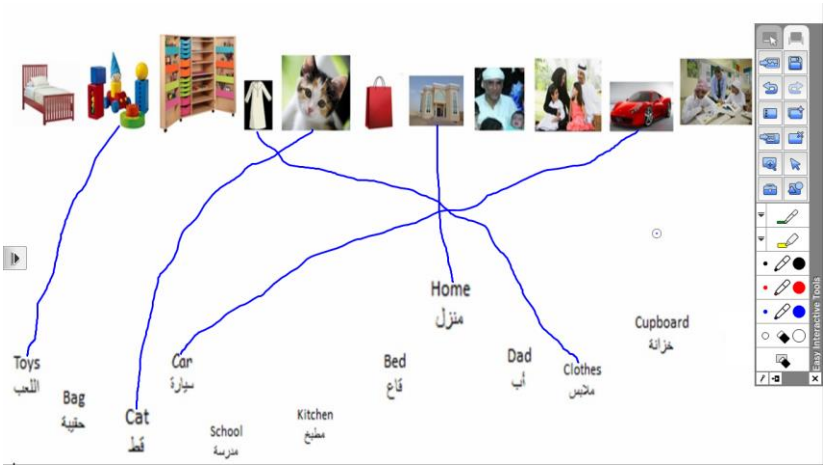

Figure D-3. Whiteboard mode in Easy Interactive Tools©

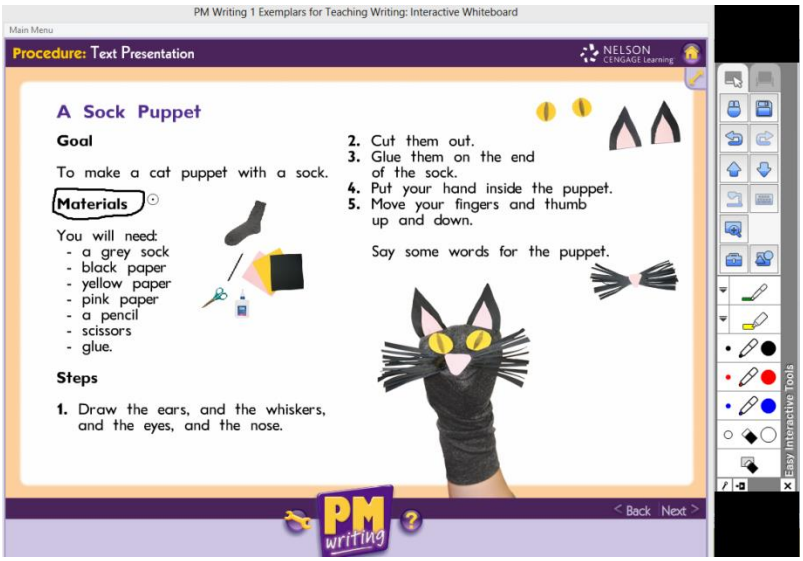

Figure D-4. Big Book Story (A Sock Puppet) with visuals and Easy Interactive Tools©

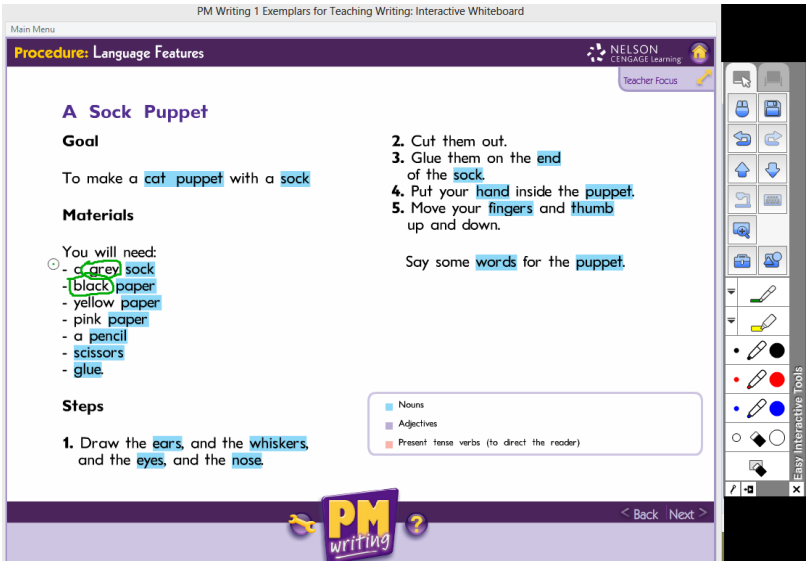

Figure D-5. Big Book Story with text highlighting features and Easy Interactive Tools © buttons

Figure D-6. Sample screen shot of Microsoft PowerPoint presentation

Figure D-7. Starting screen for Memory Game Maker©

Figure D-8. Game screen for Memory Game Maker©

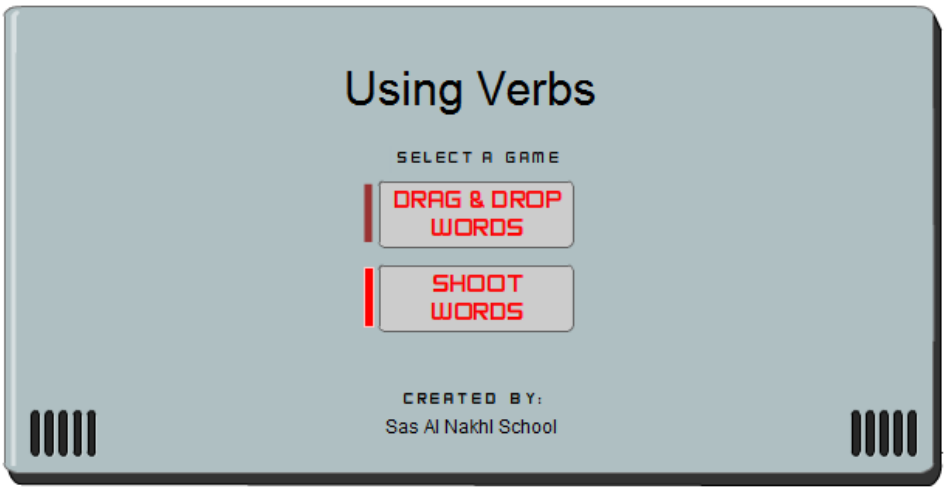

Figure D-9. Starting screen for Sentence Game Maker©

Figure D-10. Game screen for Sentence Game Maker©

Figure D-11. First six PowerPoint slides in English and Arabic

Explanation

Activity 2

Exemplar

Name: _____ Class: _____ Date: _____

How a Sock Puppet Works

Write the correct nouns in each sentence.

A puppeteer works a sock puppet.

First, the puppeteer puts his _____

inside the sock _____.

His fingers and _____ go to the end

of the sock to make the _____.

mouth

puppet

hand

thumb

Then the puppeteer says the _____

for the puppet.

He moves his _____ up and down

so that it looks like the _____

is talking.

A sock puppet needs a _____

to make it move and talk.

hand

puppeteer

words

puppet

Purpose: To use nouns correctly in sentences.

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PM writing 1

Figure D-12. Practice Sheet for “How a Sock Puppet Works?”

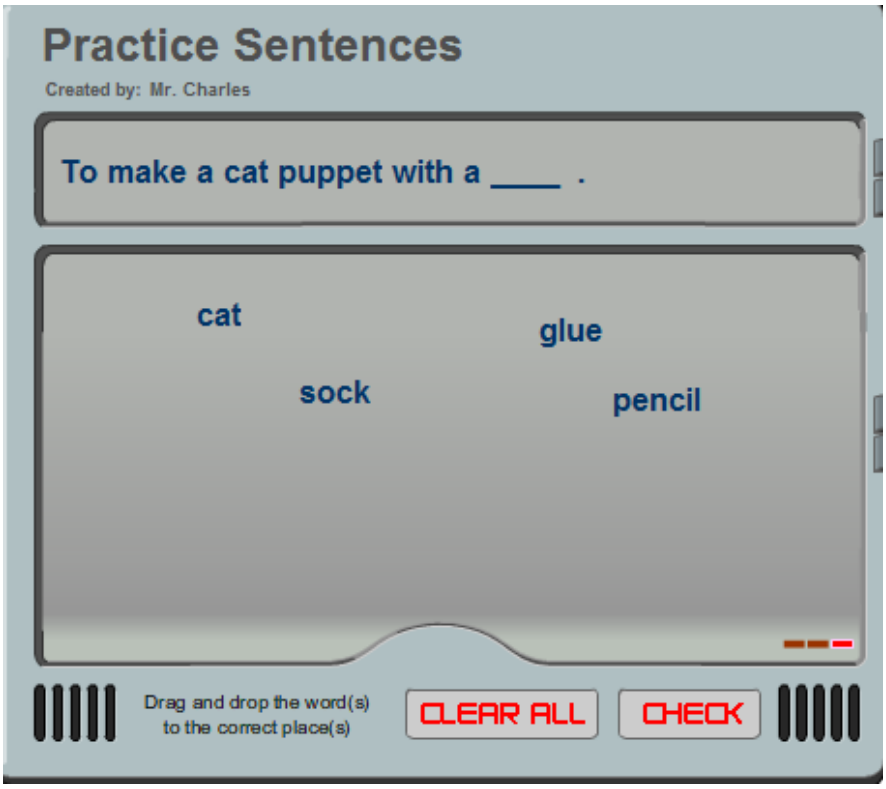

Figure D-13. Sentence Matching Game for “Sock Puppet”

Figure D-14. Matching activity for the “–ing” and “–ug” word families

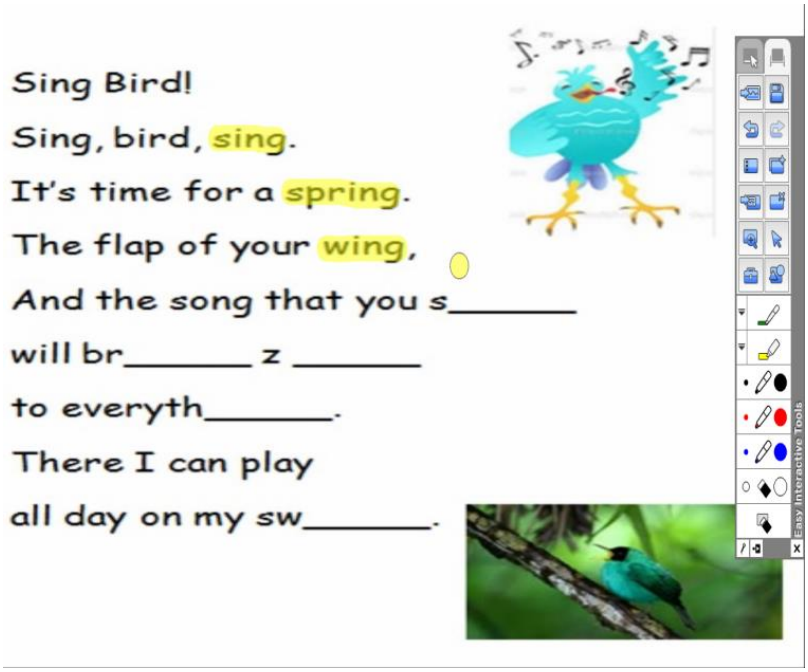

Figure D-15. “Sing Bird” Poem

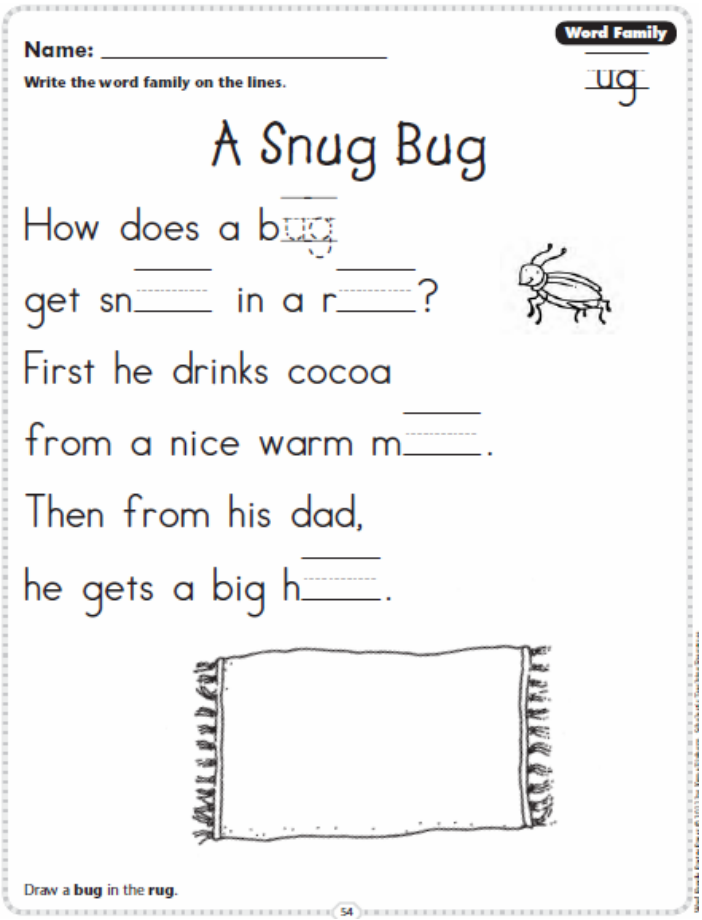

Figure D-16. “A Snug Bug” Poem

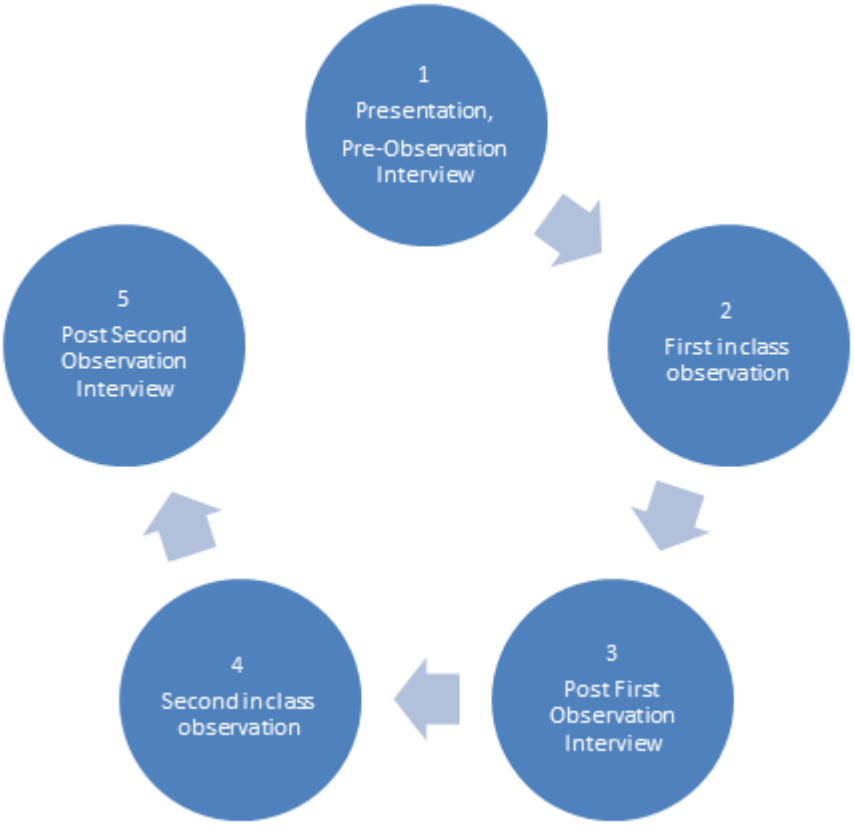

Figure D-17. Cycle of interviews, observations, and discussions

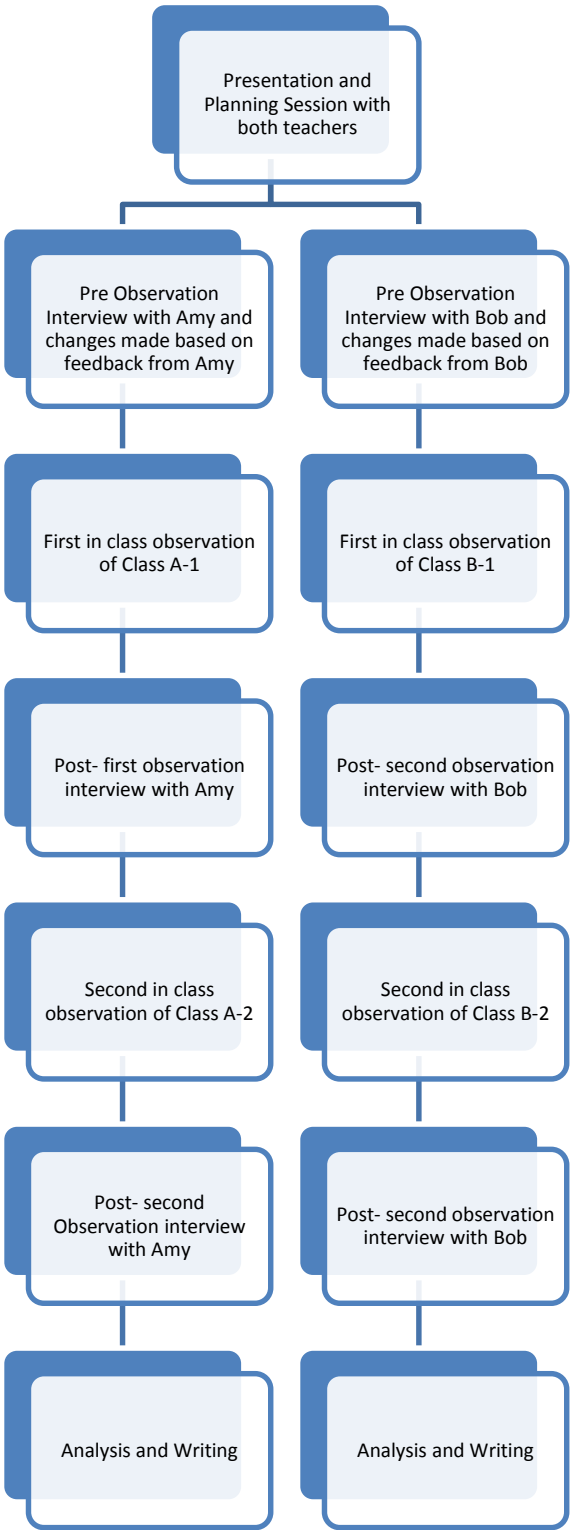

Figure D-18. Diagram of the research design

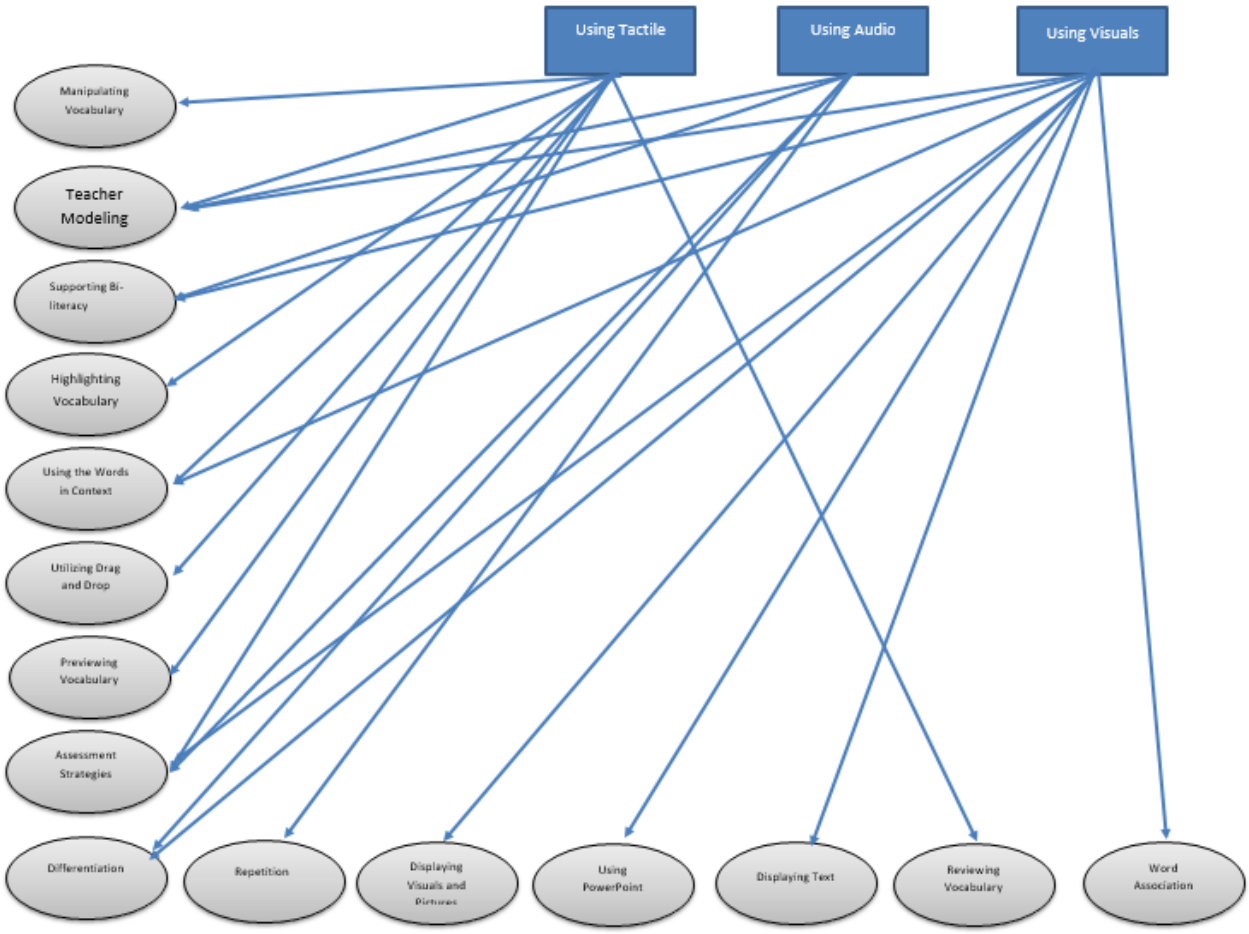

Figure D-19. Themes and corresponding codes

APPENDIX E
OBSERVATIONS AND SUGGESTIONS

The Classroom Observation Tool (Appendix B) combined parts of the Sheltered Instruction Observation Protocol (SIOP) and the International Society for Technology in Education Classroom Observation Tool (ICOT) (Echevarria, Vogt, & Short, 2000; ISTE, 2013), as described in Chapter 3. During the first in-class observation, I focused on the IWB instruction in whole group, mini lesson, small group instruction and closing. I also focused on the student interaction with language, content, application and practice.

I observed Amy on March 3 for one hour during her first and second periods using the observation protocol and taking detailed notes. Amy had 24 students in her classroom. Her room was set up with three large groups. The students started out by sitting on a large carpet in the front of the room. The IWB was located at the front of the room by the side of the teacher's desk. There were 24 local students, and her class started off with 18 students. The rest of the students came in later.

The IWB was used three separate times during the lesson. The analysis of the classroom observation is presented here based on the three times that the IWB was used for vocabulary development during the lesson. The analysis of the classroom observation revealed that the IWB was being used many different ways for vocabulary development (Table D-3).

Amy started the lesson by displaying the first part of the PowerPoint, which displayed a picture of the sock puppet. Amy then proceeded to ask the students several questions such as: "Who knows what a sock puppet is?" and "What type of materials do you need to make a sock puppet?" The students responded with what they thought the sock puppet needed, such as "scissors", "eyes", and "glue" or "colored paper". She used

this time to pre-teach the vocabulary to the students by displaying it on the interactive whiteboard with PowerPoint. She also used questioning techniques to teach the students how to say and use the words. After completing this part of the lesson, Amy displayed the PowerPoint to the students. Amy explicitly told the students that if they knew the word in Arabic that they could say it before the Arabic speech began. The Arabic words and speech were delayed. Amy asked questions about each of the vocabulary words that the students spoke. She asked the students questions such as "Who knows the Arabic word for ___?" The speech was delayed as the teacher went through each of the vocabulary words with the students. As the teacher went through the PowerPoint, the students began to speak and respond more. One of the words that the students did not know was "whiskers." They all paid attention as the word went on to the board.

After completing this part of the lesson, she assisted the students with creating the sock puppet by explicitly asking them to highlight words using the interactive whiteboard markers. Amy displayed the instructions for "making a sock puppet" and explicitly asked the students if they could highlight some of the words that were just discussed. She also explained that there were special instructions on how to make a sock puppet. The students then came up and "highlighted the special words" on how to make a sock puppet. The first student came up and highlighted "hand." All of the students had their hands up to come to the board to highlight the words. Even if they did not know the words they still wanted to come up and participate. When they highlighted the word "whiskers," the teacher prompted them to repeat the word. As they highlighted the words, the teacher also had them say the word in Arabic and English. The students

found all of the words that were required of them. After they completed the mini lesson, the teacher instructed them that they were going to go back and create sock puppets. The teacher reviewed the vocabulary with the students before they went back to their desks with a read aloud. The instructions that were on the board were also printed on paper for the students to use.

During the mini lesson, the students and Amy used the IWB by displaying the PowerPoint and vocabulary pictures. After each slide, Amy questioned the students on vocabulary that was displayed. After going over the PowerPoint, the procedural text was displayed. Several students were required to come up to the board to use the IWB marker to highlight vocabulary for the lesson.

During small group instruction, Amy split the students into separate groups. One group used the IWB while the other group worked on creating sock puppets. One group went back to create its sock puppets and the other group stayed on the carpet to play the memory game. The students in this group were split into two groups to play the game against each other. The teacher then jogged the students' memories to see if they could remember some of the words that were being discussed such as "sock" and "glue." The students were pretty excited to see if they could say the words. They had to "pick one at the top and one at the bottom," which were the teacher instructions. If they picked the wrong answer, then another student would get a chance to come up to the board to try to see if they could pick the right answer. Amy had the students practice saying the words as they picked them in the memory game. The students were excited when they picked the right answer. She reminded them that when they see the word

then they need to say it. They also needed to say the word when they saw the picture. On the last word, many of the students predicted which word would come up.

During the small group activity, the students and Amy engaged with the IWB in several ways. The students used the IWB markers to play a competitive memory game. They also practiced and predicted vocabulary. When the students made a mistake in the game, the IWB activity assisted them with self-correcting their error.

Amy used the IWB to review the vocabulary that the students went over from the lesson. The students were split into separate groups and given a fill in the blank drag and drop review game to complete. The teacher then required the students in the small group to read the sentences and drag the words into the correct spot. The game then allowed the students to check to see if they were correct. The students were encouraged to self-correct, and the program assisted with this.

The IWB was used in two ways. The IWB marker was used by the students to participate in a drag and drop activity. The drag and drop activity was also used to check students learning and understanding of vocabulary.

Amy's Suggestions for Lesson Improvement

At the end I asked her for feedback on the IWB activities and lesson. Her primary feedback was that she would like to make the lesson more challenging by getting the students to create their own sentences rather than just completing the drag and drop activity. Introducing new words confused some of the students, so she wanted them to use the words in context and sentences. She stated, "Give me a sentence that we can use with scissors in order to make a puppet. You can see that they can use that vocabulary in the sentence. They can make simple and complex sentences with the

word. The other children could support them, and we can work together." For the follow up observation, she wanted the students to create their own simple and complex sentences that we implemented.

Second Classroom Observation for Amy

I observed Amy on March 26 during her first and second periods. This was her other class that was not the same as the previous one. Twenty-four of her students were in attendance. The room was set up with three large groups. The students all started class by sitting on a large carpet in the front of the room. The IWB was located at the front of the room by the side of the teacher's desk. During the second observation, many of the interactions that I observed with the teacher and students were very similar to the first observation. One of the areas that demonstrated student understanding and vocabulary development was the closing part of the lesson when students created their own simple and complex sentences. The students created sentences using PowerPoint, the interactive whiteboard markers and the drag and drop tool of the interactive whiteboard software. The sentences started out easy and became more difficult as the students progressed.

During the mini-lesson, the area that she decided to change was having the students read the words out loud in English and Arabic. During the lesson, the students were very engaged, and they corrected Arabic when they believed that the word was wrong, which demonstrated that they understood the vocabulary. Amy pre-taught the vocabulary. She decided to change the strategy and have the students read the word and say it in Arabic and English. She told the students to remember the words and say the words in English and Arabic. All of the students were engaged. They corrected the

Arabic if there was a mistake. For instance, the word for “sock” was incorrect, so the students said it louder in Arabic and English. The students were very vocal when listening to and reading the PowerPoint. There were multiple words in Arabic for the English words, and many of the students repeated those words.

During the mini lesson, the students also came up and highlighted the words like in the first observation. One of the differences is that Amy repeated the word as the students highlighted. She asked the students if they remembered all of the words that they saw. She explained to the students to highlight the words to make the sock puppet. The emphasis was on the directions. She then read through the story and directions once and then had the students come up and highlight what they did. The goal is to “make a cat puppet with a sock.” The students then came up and highlighted many of the words.

During the small group instruction, the students played a memory game that allowed them to review their retention of the vocabulary. The students were split into separate groups. Amy explicitly stated that the one group would go back and make their sock puppets, whereas the other group stayed on the carpet to pay the memory game. Amy checked the student’s memory to see if they could remember some of the words. All of the students were engaged.

During the closing, the drag and drop activity was replaced with an activity to make their own sentences. Amy instructed the students that they were going to drag the words to create a sentence and then after they were going to read the sentences out loud. They then dragged the sentences on the line. After they dragged the first sentence they read the sentence “Draw the ears”. The next student came up and put the words in

order. The students used the IWB markers to place the words in correct order. The students in the class were noticeably paying attention to the lesson and activity. The students were then given long sentences. When the students made a mistake they corrected their own mistake. While the third student was making the sentence, the other students were completely quiet. There was not a period at the end of the sentence, and the students spotted the correction.

First Classroom Observation for Bob

The first classroom observation with Bob lasted approximately 60 minutes during first and second periods. Bob's classroom was set up with four large groups. The students start class by sitting at their desks. The IWB was located at the front of the room by the side of the teacher's desk. Bob also used the IWB for classroom management where the students could come up and mark their names for positive and negative reinforcement. The name of the program that he used for this was Class Dojo.

Analysis of the classroom observations revealed Bob used the interactive whiteboard for vocabulary development. The screenshots focused on using the highlighting aspect of the IWB as well as playing the memory game. The mini lesson lasted approximately 10 minutes. Bob started the lesson by describing what the students would be learning during the lesson and displaying the word families on the interactive whiteboard. He started the lessons by displaying a text of the poem that required the students to use the word families and by reading the first sentence of the poem "Sing Bird." He invited the students to come up to the front to highlight the word family "–ing," but he first modeled how to write the word using the IWB marker. The first word that he wrote was "sing." He repeated the word several times to the students. The

next word that he wrote was “spring.” The students were very attentive and several students predicted out loud the next word that they think thought he should highlight. He then asked for one volunteer to come up to the board and highlight the next word family. The majority of the students raised their hands.

The next student came up and successfully highlighted the word. Bob then asked the students if it was correct and they all said that it was correct. Bob then continued and asked for another volunteer. As he read the poem, the students completed the poem both out loud and on the interactive whiteboard. They completed the next three sentences of the word family poem. Each of the students successfully came up and wrote their words. One of the issues that the students had was the height of the board and their pushing down with the marker to get the marker to successfully write. He mentioned this during the pre-interview as a challenge of using the IWB. As they reached the conclusion of the mini lesson, Bob then asked the students to read the poem together, and as he reached the end of each sentence, he raised his voice to bring out the word family. He purposely left the last sentence blank so that the students could predict the word family at the end, as well as the sound of the word. Before going into small group instruction, he asked the students if they could say some of the words that they went over in the poem. Some of the students said words that had the word family “–ing” but were not in the poem.

Bob then split the students into separate groups. Each of the groups completed a word family activity sheet while a smaller group was pulled to the board to complete a small group activity. The activity that they completed required them to use the interactive whiteboard tools to place a circle and a triangle around the word families. He

invited the first group of three students up to the board. They placed a triangle around the “–ing” word families and a circle around the “–ug.” The first student in the group came up and successfully circled the first word “rug” for the word family “–ug.” Bob then asked the next student to find a word with the word family “–ing” and to place a triangle around it. Each successive student had no problems circling and placing triangles around the word families that they were assigned.

For the closing part of the lesson Bob placed the story “A Snug Bug” on the board. The students were then required to come up to the board and write the ending of the words using the IWB marker. Bob invited one students to come up and complete the first sentence, “How does a bug get snug in a rug?” The rest of students successfully came up and wrote the words using the IWB marker.

Suggested Improvements from Bob

I met with Bob about one week after the first observation. He gave some feedback on the IWB activities and lesson. Bob felt that usage of the interactive whiteboard could have been improved by making the activity more accessible to smaller students. He stated that he would like for this to change in the follow-up lesson. He also felt that the circling activity was a good activity for students, although he would like to see a different format. He stated, “There could have been different types of families in the sentence. Overall I think it was effective. Things could always have been improved.” One of the challenges for the boys was the delay between them pressing on the board with the interactive whiteboard marker.

Second Classroom Observation for Bob

During Bob’s second classroom observation, I made two changes to the lesson from the feedback that I received during the interview. I redesigned the PowerPoint and

IWB activity during the three parts of the lesson (whole group, small group, and closing) to make it more accessible to smaller students. I also included an additional word family during the small group activity to accommodate students that needed a more challenging activity. We also included different types of word families, specifically adding “–at” to the small group activity. The second classroom observation with Bob lasted approximately 60 minutes during first and second periods.

Bob’s classroom was set up with four different groups. The IWB was placed near the front of the room on the side of the teacher’s desk. Bob also used the IWB for classroom management in his second class in a similar way to how he used it in his first class.

The mini lesson lasted approximately 10 minutes. Bob began the lesson in a similar way to the previous observation by telling the students what they would be learning during the lesson. He displayed the word families on the IWB and displayed the text of the poem that they would be reading together. In the classroom, all of the students were attentive. He read the first sentence of the poem “Sing Bird,” and he modeled how to write the word family, which occurred at the end of the sentence. He invited the first students to come up to the board to write the first word. After he invited several students up to the board to write the endings of the word families, they read the entire poem aloud. Bob emphasized the endings of each of the words. He then asked the students to come up with words that are not in the poem but that used the word family. Three students raised their hands and came up with several words that were not in the poem.

Bob split the students into separate groups. The other groups that were not participating in the interactive whiteboard activity completed a word family activity sheet. The activity included an additional word family that required the students to place a diamond around the words that included the word family “–at” in addition to the other shapes placed around the “–ing” and “–ug” word families. The first student in the group came up and successfully circled the first word “hug” for the word family “–ug.” Another student came up to the board and circled the word “rat” but placed the wrong shape around the word. He then asked the next student to find a word with the word family “–ing” and to place a triangle around it. He assisted the student by helping the student identify his mistake. The student corrected his mistake and successfully placed the shapes around the next words.

Bob placed the story “A Snug Bug” on the IWB. He then invited several students up to the board to write the endings of the words using the IWB marker. The students had no problem using the IWB marker to write the rest of the ends of the sentences. The students seemed to grasp the concept of word families in this particular activity.

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BIOGRAPHICAL SKETCH

Charles Martin is an educator, and his experiences include teaching in the United States and the United Arab Emirates in elementary settings. Charles uses technology-based applications and instructional design best practices to guide his efforts with teaching English as a second language to students. He holds a Bachelor of Arts (2005) and Master of Education (2006) from Georgia State University. He received his Ed.D. from the University of Florida in the summer of 2015. His research interests include Technology Enhanced Language Learning software, teaching English as a second language and utilizing interactive whiteboards.